

CONNECTED PSEUDOSPECTRA OF A NONNORMAL TOEPLITZ PATH: THE OPEN HATANO–NELSON CHAIN*

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Abstract. For the nonnormal tridiagonal Toeplitz matrix $A_n(a) = \text{tridiag}(a, 0, 1)$, $0 < a < 1$, we determine the finite-size transition at which its Euclidean ε -pseudospectrum becomes connected. The least singular value of $(x + iy)I_n - A_n(a)$ is nondecreasing in $|y|$, so every pseudospectral component retracts onto an interval of the real axis and is therefore contractible. If γ_n is the largest least-singular-value barrier between adjacent open-boundary eigenvalues, then connectedness is equivalent to $\varepsilon > \gamma_n$. We prove the strict finite-size comparison $\gamma_{n+1} < \gamma_n$ and sharp-order bounds $\gamma_n \asymp_a (n+1)a^{n/2}$. Consequently, the first connected matrix size is also the eventual connectedness threshold, and, for fixed a , it equals $2(\log(1/\varepsilon) + \log\log(1/\varepsilon))/\log(1/a) + O_a(1)$ as ε tends to zero. In the open-boundary Hatano–Nelson model, the coefficient $2/\log(1/a)$ is the right-eigenmode amplitude skin depth, in lattice-spacing units; the threshold describes merger of worst-case spectral-uncertainty clouds at a prescribed perturbation level.

Key words. pseudospectrum, nonnormal Toeplitz matrix, Hatano–Nelson model, non-Hermitian skin effect, least singular value, Chebyshev polynomial

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1. Introduction. For a matrix $A \in \mathbb{C}^{n \times n}$, the Euclidean ε -pseudospectrum is

$$(1.1) \quad \sigma_\varepsilon(A) = \sigma(A) \cup \{z \in \mathbb{C} : \|(zI_n - A)^{-1}\|_2 > \varepsilon^{-1}\} = \{z \in \mathbb{C} : s_{\min}(zI_n - A) < \varepsilon\}.$$

Equivalently, it is the union of the spectra of $A + \Delta A$ over all unstructured complex perturbations $\Delta A \in \mathbb{C}^{n \times n}$ with $\|\Delta A\|_2 < \varepsilon$; see [26]. For nonnormal matrices, these sets reveal spectral sensitivity and resolvent growth that the eigenvalues alone can miss; see [24, 26]. Classical spectral limit sets for banded Toeplitz sections go back to Schmidt and Spitzer [22]. Finite Toeplitz matrices are a setting in which the distinction between spectra and pseudospectra can be exponentially large in the dimension; their finite-section resolvents and pseudospectra have been studied in [6, 7, 8, 21].

We study the real tridiagonal Toeplitz path

$$(1.2) \quad A_n(a) = \text{tridiag}(a, 0, 1) = \begin{bmatrix} 0 & 1 & & \\ a & 0 & \ddots & \\ & \ddots & \ddots & 1 \\ & & a & 0 \end{bmatrix}, \quad n \geq 2, \quad 0 < a < 1,$$

where the three arguments of tridiag denote the subdiagonal, diagonal, and super-diagonal, respectively. Closed-form spectral, conditioning, and pseudospectral results for general tridiagonal Toeplitz matrices are developed in [17]. Although $A_n(a)$ is diagonally similar to a symmetric Jacobi matrix, the similarity is not unitary, and hence it does not preserve the Euclidean pseudospectrum. Our question is finite-dimensional and topological: as n grows at fixed a and ε , can connectedness appear and then disappear, or is there a unique transition?

The same family is the clean open-boundary Hatano–Nelson chain [12, 13]. With the site convention in which t_R moves a particle from j to $j+1$ and t_L moves it from

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38 $j + 1$ to j , assume $0 < t_R < t_L$ and write

$$39 \quad H_n^{\text{OBC}} = t_L A_n(a), \quad a = \frac{t_R}{t_L}.$$

40 The dimensionless logarithmic parametrization used in the proof is

$$41 \quad (1.3) \quad h = -\frac{1}{2} \log a > 0, \quad a = e^{-2h}, \quad r = \sqrt{a} = e^{-h}.$$

42 The diagonal gauge

$$43 \quad \mathcal{G}_n(\varrho) = \text{diag}(1, \varrho, \dots, \varrho^{n-1})$$

44 and the upper-shift matrix $V_n = \text{tridiag}(0, 0, 1)$ give the dimensionless similarity

$$45 \quad (1.4) \quad A_n(a) = \mathcal{G}_n(r) [r(V_n + V_n^T)] \mathcal{G}_n(r)^{-1}.$$

46 Multiplying by t_L and using $t_L r = \sqrt{t_L t_R}$ recovers the usual physical imaginary-gauge
 47 formula for H_n^{OBC} . Thus the open-boundary eigenvalues are real, while the right
 48 eigenmodes have the envelope $e^{-h(j-1)}$ and accumulate at the left edge. This is the
 49 non-Hermitian skin effect [18, 27, 28]; see [29] for a review. Under periodic boundary
 50 conditions the Bloch curve is the ellipse

$$51 \quad \mathcal{E}_{\text{PBC}}(k) = t_L e^{ik} + t_R e^{-ik} = (t_L + t_R) \cos k + i(t_L - t_R) \sin k.$$

52 Early work connecting Hatano–Nelson–type localization with pseudospectra treated
 53 random bidiagonal and Toeplitz models [25]. Direct pseudospectral interpretations
 54 of nonnormal boundary sensitivity appear in [19, 20]. Mathematical foundations for
 55 the skin effect were developed in [2], while spectra and pseudospectra of tridiagonal k -
 56 Toeplitz matrices were treated in [4, 5]. Pseudospectral scaling in asymmetric lattices
 57 was studied in [15]; see also the preprint [23] for a closely related pseudospectral
 58 analysis of the open Hatano–Nelson chain.

59 For general matrices, the geometry and coalescence of pseudospectral components
 60 have been studied by variational and distance-to-defectivity methods [1, 9, 16]. The
 61 present work complements that theory by determining, for the clean scalar tridiagonal
 62 Toeplitz path, an exact finite-dimensional connectedness criterion, contractibility
 63 of every component, and a strict comparison of the merger barrier in consecutive
 64 dimensions.

65 Our contributions are as follows. First, we prove vertical monotonicity of the
 66 least singular value, which reduces planar topology to a real-axis barrier problem and
 67 shows, more strongly, that every pseudospectral component is contractible. Second, a
 68 signed persymmetric pencil and a folded Chebyshev parametrization give the strict
 69 comparison of the largest barrier in consecutive dimensions. Third, rectangular
 70 factorizations give sharp-order two-sided bounds and the resulting Lambert- W finite-
 71 size law. Connectedness concerns the union over all *unstructured* perturbations: it does
 72 not imply that any single perturbation supplies eigenvalues throughout a connecting
 73 bridge, nor does it describe the attainable set for a prescribed structured perturbation
 74 family. The transition is also not the onset of the skin effect, which is already present
 75 in (1.4).

76 The paper is organized as follows. Section 2 states the main result. Vertical
 77 monotonicity is proved in section 3; the strict barrier comparison occupies section 4;
 78 and the topological criterion and quantitative bounds are established in section 5.
 79 Numerical illustrations and the physical interpretation are given in sections 6 and 7.

2. Model and main results. For $n \geq 2$ and $0 < a < 1$, let

$$A_n(a) = \text{tridiag}(a, 0, 1).$$

When a is fixed, we write A_n for $A_n(a)$. For $\varepsilon > 0$, write

$$\sigma_\varepsilon(A_n) = \{z \in \mathbb{C} : s_{\min}(zI_n - A_n) < \varepsilon\}.$$

Define

$$N_e(a, \varepsilon) = \min\{N \geq 2 : \sigma_\varepsilon(A_n) \text{ is connected for every } n \geq N\},$$

$$N_f(a, \varepsilon) = \min\{N \geq 2 : \sigma_\varepsilon(A_N) \text{ is connected}\}.$$

Throughout the remainder of the article, put

$$r = \sqrt{a}, \quad J_n = [\mathbf{1}_{\{i+j=n+1\}}]_{i,j=1}^n.$$

Thus $A_n = V_n + aV_n^T$. Throughout, $\mathbf{e}_j$ denotes a Euclidean coordinate vector of the ambient dimension, and $U_j(t)$ denotes the Chebyshev polynomial of the second kind, with $U_{-1} = 0$.

When the dimension is n , write $N = n + 1$ and define

$$(2.1) \quad I_n^{\text{sp}} = \left[-2r \cos \frac{\pi}{N}, 2r \cos \frac{\pi}{N}\right], \quad g_n(x) = s_{\min}(xI_n - A_n), \quad \gamma_n = \max_{x \in I_n^{\text{sp}}} g_n(x).$$

The explicit lower and upper barriers used below are

$$(2.2) \quad \mathcal{U}_n(a) = r^n \csc \frac{\pi}{N},$$

$$(2.3) \quad \mathcal{L}_n(a) = \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right) \frac{r^n(1 - r^2)}{(1 - r^{2N}) \sin \frac{3\pi}{2N}}.$$

THEOREM 2.1 (Main theorem). *For every $0 < a < 1$ and $\varepsilon > 0$, the following assertions hold.*

1. *For every $n \geq 2$, every connected component of $\sigma_\varepsilon(A_n(a))$ is contractible.*
2. *For $n \geq 2$, the barrier sequence is strictly decreasing, $\gamma_{n+1} < \gamma_n$, tends to zero, and*

$$\sigma_\varepsilon(A_n(a)) \text{ is connected} \iff \varepsilon > \gamma_n.$$

3. *The first-hit and eventual connectedness thresholds coincide:*

$$N_c(a, \varepsilon) := N_f(a, \varepsilon) = N_e(a, \varepsilon).$$

Moreover,

$$(2.4) \quad 1 + \max\{n \geq 2 : \mathcal{L}_n(a) \geq \varepsilon\} \leq N_c(a, \varepsilon) \leq \min\{n \geq 2 : \mathcal{U}_n(a) < \varepsilon\},$$

where the maximum of the empty set is understood to be 1.

4. *For fixed a , as $\varepsilon \downarrow 0$,*

$$(2.5) \quad N_c(a, \varepsilon) = \frac{2}{\log(1/a)} (\log(1/\varepsilon) + \log \log(1/\varepsilon)) + O_a(1),$$

and $N_c(a, \varepsilon) = 2$ if and only if $\varepsilon > a$.

3. Vertical monotonicity. We first prove the vertical monotonicity that will be used in the topological part of the argument.

LEMMA 3.1 (Vertical monotonicity). *For every real x ,*

$$y \mapsto s_{\min}((x + iy)I_n - A_n), \quad y \geq 0,$$

is nondecreasing.

Proof. Put $Y = y^2$, and, for $t \geq 0$, define

$$\mathcal{P}_n(Y, t) = ((x + i\sqrt{Y})I_n - A_n)^*((x + i\sqrt{Y})I_n - A_n) - t^2 I_n, \quad F_n(Y, t) = \det \mathcal{P}_n(Y, t).$$

The same definition is used in every dimension: for $k \geq 1$, $\mathcal{P}_k(Y, t)$ is obtained by replacing n by k , $F_k(Y, t) = \det \mathcal{P}_k(Y, t)$, and $F_0 = 1$. We shall prove the finite-dimensional implication

$$(3.1) \quad \mathcal{P}_n(Y, t) > 0 \implies \partial_Y F_n(Y, t) > 0.$$

At a fixed value of Y , write

$$d = x^2 + Y + 1 + a^2 - t^2, \quad b = -(1 + a)x + i(1 - a)\sqrt{Y}.$$

For $n \geq 2$, the matrix $\mathcal{P}_n(Y, t)$ has diagonal

$$d - 1, d, \dots, d, d - a^2,$$

whereas for $n = 1$ its sole entry is $d - 1 - a^2$. In all dimensions its first upper diagonal is b , its first lower diagonal is $\bar{b}$, and its second off-diagonals are a . Let T_k be the determinant of the pure pentadiagonal Toeplitz matrix with diagonal d and these two off-diagonals. Set $T_{-1} = 0$, $T_0 = 1$, and put

$$(3.2) \quad D_k = T_k - T_{k-1}, \quad R_k = T_k - a^2 T_{k-1}.$$

Indeed, a determinant is affine in each diagonal entry. Replacing the first diagonal entry d by $d - 1$ subtracts the cofactor T_{k-1} , whereas replacing the last one by $d - a^2$ subtracts $a^2 T_{k-1}$. Thus, for $k < n$, D_k and R_k are, respectively, the leading and trailing principal minors of $\mathcal{P}_n(Y, t)$.

We record the continuant calculation that contains all the dependence on Y . With $T_k = 0$ for $k < 0$, a Laplace expansion over the last two rows, grouped according to the largest displaced index in the contributing permutation, gives for every $k \geq 1$

$$(3.3) \quad \begin{aligned} T_k &= dT_{k-1} - (|b|^2 - a^2)T_{k-2} - (2a^2d - a(b^2 + \bar{b}^2))T_{k-3} \\ &\quad - a^2(|b|^2 - a^2)T_{k-4} + a^4dT_{k-5} - a^6T_{k-6} - a^2\mathbf{1}_{\{k=2\}}. \end{aligned}$$

For example, $T_1 = d$ and $T_2 = d^2 - |b|^2$; these also verify the only inhomogeneous coefficient in (3.3). Multiplying the recurrence by z^k and summing gives

$$(3.4) \quad \sum_{k \geq 0} T_k z^k = \frac{1 - a^2 z^2}{Q(z, Y)},$$

where

$$\begin{aligned} Q(z, Y) &= 1 - dz + (|b|^2 - a^2)z^2 + (2a^2d - a(b^2 + \bar{b}^2))z^3 \\ &\quad + a^2(|b|^2 - a^2)z^4 - a^4dz^5 + a^6z^6. \end{aligned}$$

Consequently, with $\mathcal{D}(z, Y) = \sum_{k \geq 0} D_k z^k$,

$$(3.5) \quad \mathcal{D}(z, Y) = \frac{(1-z)(1-a^2 z^2)}{Q(z, Y)}, \quad \sum_{k \geq 0} F_k(Y, t) z^k = (1-a^2 z) \mathcal{D}(z, Y).$$

For the second identity, first change the initial diagonal, producing D_k , and then change the final diagonal; its cofactor is D_{k-1} . Thus $F_k = D_k - a^2 D_{k-1}$, which proves the asserted generating function. Direct substitution of d and b gives

$$\partial_Y Q = -z(1-z)(1+az)^2(1-a^2 z).$$

Differentiating (3.5) therefore yields

$$(3.6) \quad \sum_{k \geq 0} \partial_Y F_k(Y, t) z^k = \varphi_a(z) \mathcal{D}(z, Y)^2, \quad \varphi_a(z) = \frac{z(1+az)(1-a^2 z)^2}{1-az}.$$

Suppose that $\mathcal{P}_n(Y, t) > 0$. Then

$$(3.7) \quad D_k > 0, \quad R_k > 0 \quad (0 \leq k < n).$$

Put $u_k = \sum_{j=0}^k D_j D_{k-j}$, with $u_k = 0$ for $k < 0$. The coefficients of φ_a , beginning with the coefficient of z , are

$$1, \quad 2a(1-a), \quad a^2(a^2-4a+2), \quad 2a^{k-1}(1-a)^2 \quad (k \geq 4).$$

Thus

$$(3.8) \quad \begin{aligned} \partial_Y F_n &= u_{n-1} + 2a(1-a)u_{n-2} + a^2(a^2-4a+2)u_{n-3} \\ &\quad + 2(1-a)^2 \sum_{k=4}^n a^{k-1} u_{n-k}. \end{aligned}$$

Only the third coefficient can be negative. We now give the cofactor inequality that absorbs it. Let $j, k \geq 1$, $j+k = n-1$. Deleting index $k+1$ from $\mathcal{P}_n$ leaves blocks of sizes k and j , coupled only by the second off-diagonal a . In block form, the deleted principal submatrix is

$$(3.9) \quad \begin{bmatrix} L_k & a\mathbf{e}_k \mathbf{e}_1^T \\ a\mathbf{e}_1 \mathbf{e}_k^T & R_j^{\text{mat}} \end{bmatrix}, \quad \det L_k = D_k, \quad \det R_j^{\text{mat}} = R_j.$$

Expanding once in each of the two coupling entries shows that its determinant is

$$(3.9) \quad D_k R_j - a^2 D_{k-1} R_{j-1} > 0.$$

Deleting index $j+1$ instead gives the same inequality with j, k interchanged. Define

$$\vartheta_k = \frac{D_k}{D_{k-1}}, \quad \varpi_k = \frac{R_k}{R_{k-1}}.$$

Since $T_k = \sum_{\ell=0}^k D_\ell$, (3.2) gives

$$(3.10) \quad R_k = D_k + (1-a^2) \sum_{\ell=0}^{k-1} D_\ell, \quad R_k - R_{k-1} = D_k - a^2 D_{k-1}.$$

Hence $\vartheta_k \leq a^2$ implies $\varpi_k \leq 1$. On dividing (3.9) and its companion by positive factors, we obtain

$$(3.11) \quad \vartheta_k \varpi_j > a^2, \quad \vartheta_j \varpi_k > a^2.$$

If both ϑ_j and ϑ_k were at most a^2 , then both ϖ_j and ϖ_k would be at most 1, contradicting (3.11). Hence, after interchanging j, k if necessary, $\vartheta_j > a^2$. Moreover,

$$\vartheta_j R_{j-1} - a^2 R_j = (\vartheta_j - a^2)(R_{j-1} - a^2 D_{j-1}),$$

and the second factor equals $(1 - a^2) \sum_{\ell=0}^{j-1} D_\ell > 0$. Therefore

$$\vartheta_j > a^2 \implies \varpi_j < \vartheta_j / a^2.$$

The first inequality in (3.11) now gives $a^2 < \vartheta_k \varpi_j < \vartheta_j \vartheta_k / a^2$, and therefore

$$(3.12) \quad \vartheta_j \vartheta_k > a^4.$$

Consequently, for $n \geq 3$, applying (3.12) to $j = p + 1$, $k = n - 2 - p$ gives

$$D_{p+1} D_{n-2-p} > a^4 D_p D_{n-3-p}, \quad 0 \leq p \leq n - 3.$$

Summing and then adding the two positive endpoint terms gives

$$(3.13) \quad u_{n-1} = 2D_{n-1} + \sum_{j=1}^{n-2} D_j D_{n-1-j} > a^4 u_{n-3}.$$

If the third coefficient in (3.8) is negative, write it as $-\mathcal{B}_{\text{neg}}$, where $\mathcal{B}_{\text{neg}} = a^2(4a - 2 - a^2)$. Since

$$a^4 - \mathcal{B}_{\text{neg}} = 2a^2(1 - a)^2 > 0,$$

equations (3.8) and (3.13) prove (3.1). If that coefficient is nonnegative, the implication is immediate. For $n = 1$, one has $\partial_Y F_1 = 1$; for $n = 2$, one has $\partial_Y F_2 = 2D_1 + 2a(1 - a) > 0$. Thus (3.1) holds in every dimension.

Finally fix $Y_0 \geq 0$. If the least singular value at Y_0 is zero, the desired inequality is automatic. Otherwise take $0 \leq t < s_{\min}((x + i\sqrt{Y_0})I_n - A_n)$. On the maximal interval to the right of Y_0 on which $\mathcal{P}_n(Y, t) > 0$, implication (3.1) says that its determinant is strictly increasing. By continuity of Hermitian eigenvalues, a finite right endpoint would be a first loss of positive definiteness at a positive-semidefinite singular matrix, and hence would have determinant zero. This is impossible because the determinant has increased from its positive value at Y_0 . Thus $\mathcal{P}_n(Y, t) > 0$ for every $Y \geq Y_0$. Letting t increase to the least singular value at Y_0 proves the lemma. $\square$

4. Strict finite-size comparison of real-gap barriers. We next establish the strict comparison between consecutive dimensions. Write the eigenvalues in decreasing order as

$$\lambda_{n,k} = 2r \cos \frac{k\pi}{n+1}, \quad 1 \leq k \leq n,$$

and call

$$G_{n,k} = [\lambda_{n,k+1}, \lambda_{n,k}], \quad \gamma_{n,k} = \max_{x \in G_{n,k}} g_n(x), \quad 1 \leq k < n,$$

the closed spectral gaps and their heights. Continuity gives existence of every maximum, and the spectral endpoints have height zero. Every interior gap point lies outside the

spectrum, so $xI_n - A_n$ is nonsingular there and $g_n(x) > 0$. Consequently every gap height is positive and every point at which it is attained lies in the open gap. Hence

$$(4.1) \quad \gamma_n = \max_{1 \leq k < n} \gamma_{n,k}.$$

LEMMA 4.1 (Strict barrier comparison). *For every $n \geq 2$,*

$$\gamma_{n+1} < \gamma_n.$$

We give the complete finite-path calculation. Persymmetry gives $A_n J_n = J_n A_n^T$, so $B_n(x) = (xI_n - A_n)J_n$ is real symmetric. Moreover,

$$(xI_n - A_n - sJ_n)J_n = B_n(x) - sI_n.$$

Since J_n is orthogonal, the absolute eigenvalues of $B_n(x)$ are exactly the singular values of $xI_n - A_n$, including multiplicity. Consequently,

$$(4.2) \quad s_{\min}(xI_n - A_n) = \min\{|s| : \det(xI_n - A_n - sJ_n) = 0\}.$$

Let $q_n(x, s) = \det(xI_n - A_n - sJ_n)$.

LEMMA 4.2 (The middle signed branch). *Write the eigenvalues of the real symmetric matrix*

$$B_n(x) = (xI_n - A_n)J_n$$

in decreasing order as $\alpha_1(x) \geq \dots \geq \alpha_n(x)$, and abbreviate $\lambda_k = \lambda_{n,k}$. If $\lambda_{j+1} < x < \lambda_j$, then

$$s_{\min}(xI_n - A_n) = |s_*(x)|, \quad \text{sgn}(s_*(x)) = (-1)^j,$$

where

$$s_*(x) = \begin{cases} \alpha_m(x), & n = 2m, \\ \alpha_{m+1}(x), & n = 2m + 1. \end{cases}$$

Proof. Let $A_n^{\text{sym}} = r(V_n + V_n^T)$. Since $A_n = \mathcal{G}_n(r)A_n^{\text{sym}}\mathcal{G}_n(r)^{-1}$ and $\mathcal{G}_n(r)^{-1}J_n = r^{-(n-1)}J_n\mathcal{G}_n(r)$,

$$(4.3) \quad B_n(x) = r^{-(n-1)}\mathcal{G}_n(r)(xI_n - A_n^{\text{sym}})J_n\mathcal{G}_n(r).$$

The sine eigenvector of A_n^{sym} belonging to λ_k is an eigenvector of J_n with eigenvalue $(-1)^{k+1}$. Hence (4.3) and Sylvester's law of inertia show that $B_n(x)$ has the same inertia as the diagonal matrix with entries

$$(-1)^{k+1}(x - \lambda_k), \quad 1 \leq k \leq n.$$

Counting signs gives

$$(4.4) \quad \nu_+(B_{2m}(x)) = \begin{cases} m, & j \text{ even}, \\ m-1, & j \text{ odd}, \end{cases} \quad \nu_+(B_{2m+1}(x)) = \begin{cases} m+1, & j \text{ even}, \\ m, & j \text{ odd}. \end{cases}$$

First let $n = 2m$. Permute the coordinates into the two groups

$$(1, \dots, m), \quad (2m, 2m-1, \dots, m+1).$$

238 In these coordinates

$$\begin{aligned}
 239 \quad (4.5) \quad & B_{2m}(x) = \hat{P}_m(x) - \mathcal{R}_{\text{bd}}, \\
 & \hat{P}_m(x) = \begin{bmatrix} 0 & P_m(x) \\ P_m(x)^T & 0 \end{bmatrix}, \\
 & \mathcal{R}_{\text{bd}} = \begin{bmatrix} \mathbf{e}_m \mathbf{e}_m^T & 0 \\ 0 & a \mathbf{e}_m \mathbf{e}_m^T \end{bmatrix},
 \end{aligned}$$

240 where $P_m(x) = xI_m - A_m$. Thus $\mathcal{R}_{\text{bd}} \geq 0$ and $\text{rank } \mathcal{R}_{\text{bd}} = 2$. For $\Gamma = \text{diag}(I_m, -I_m)$,

$$241 \quad \tilde{B} = -\Gamma B_{2m}(x) \Gamma = \hat{P}_m(x) + \mathcal{R}_{\text{bd}} = B_{2m}(x) + 2\mathcal{R}_{\text{bd}}.$$

242 If $\tilde{\alpha}_1 \geq \cdots \geq \tilde{\alpha}_{2m}$ are the eigenvalues of $\tilde{B}$, then

$$243 \quad (4.6) \quad \tilde{\alpha}_k = -\alpha_{2m+1-k}.$$

244 Positive-semidefinite rank-two interlacing gives

$$245 \quad (4.7) \quad \tilde{\alpha}_k \geq \alpha_k \geq \tilde{\alpha}_{k+2}, \quad 1 \leq k \leq 2m-2.$$

246 If j is even, the only candidates for minimum modulus have moduli α_m and $-\alpha_{m+1} =$
 247 $\tilde{\alpha}_m$, and $\alpha_m \leq \tilde{\alpha}_m$. If j is odd and $m \geq 2$, the two candidates have moduli α_{m-1}
 248 and $-\alpha_m = \tilde{\alpha}_{m+1}$; taking $k = m-1$ in (4.7) gives $\alpha_{m-1} \geq \tilde{\alpha}_{m+1}$. For $m = 1$,
 249 both eigenvalues in the sole gap are negative, so the larger one α_1 has minimum
 250 modulus. Thus α_m always attains the minimum modulus, and its sign follows from
 251 (4.4). Equality in an interlacing comparison is harmless: it only gives a multiple least
 252 singular value.

253 Now let $n = 2m+1$. Group the first and last m coordinates as above and leave
 254 the middle coordinate last. Then

$$255 \quad (4.8) \quad B_{2m+1}(x) = \begin{bmatrix} \hat{P}_m(x) & \mathbf{w} \\ \mathbf{w}^T & x \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} -\mathbf{e}_m \\ -a\mathbf{e}_m \end{bmatrix}.$$

256 In an open spectral gap of A_{2m+1} , $P_m(x)$ is nonsingular. Indeed, the explicit path
 257 spectrum gives

$$258 \quad \text{spec}(A_m) = \left\{ 2r \cos \frac{\ell\pi}{m+1} : 1 \leq \ell \leq m \right\} = \{ \lambda_{2m+1, 2\ell} : 1 \leq \ell \leq m \},$$

259 so its zeros are exactly the even-indexed eigenvalues of A_{2m+1} , none of which lies in an
 260 open spectral gap. The eigenvalues of the principal submatrix $\hat{P}_m(x)$ are consequently

$$261 \quad s_1(P_m) \geq \cdots \geq s_m(P_m) > 0 > -s_m(P_m) \geq \cdots \geq -s_1(P_m).$$

262 Cauchy interlacing gives

$$263 \quad \alpha_m \geq s_m(P_m) \geq \alpha_{m+1} \geq -s_m(P_m) \geq \alpha_{m+2}.$$

264 If j is even, (4.4) and interlacing give

$$265 \quad 0 < \alpha_{m+1} \leq s_m(P_m) \leq -\alpha_{m+2};$$

266 if j is odd, they give

$$267 \quad 0 < -\alpha_{m+1} \leq s_m(P_m) \leq \alpha_m.$$

Every other eigenvalue is farther from zero. Hence α_{m+1} attains the minimum modulus, with sign prescribed by (4.4); equality again only means multiplicity. At a spectral endpoint the zero is simple because A_n is similar to the Jacobi matrix A_n^{sym} , whose eigenvalues are simple. Thus the selected ordered branch extends continuously to zero at both ends of the closed gap. Finally, multiplication by J_n does not change singular values, which proves the assertion. $\square$

Proof of Lemma 4.1. Folded continuants. Put

$$C_0 = x^2 + (1-a)^2 - s^2, \quad \zeta_s = \frac{1+a^2+(1+a)s}{2a},$$

and define the folded discriminant and recurrence coefficient by

$$\begin{aligned} \Delta_{\text{fold}} &= ((1+a-x)^2 - s^2)((1+a+x)^2 - s^2), \\ \mathfrak{b} &= (1+a^2)x^2 + 2as^2 + 2a^2 - 2a - 2a^3. \end{aligned}$$

The global recurrence polynomial is (4.9)

$$\begin{aligned} \Pi(z) &:= 1 - C_0z + \mathfrak{b}z^2 - a^2C_0z^3 + a^4z^4, \\ \left. \begin{aligned} 2a(u+v) &= C_0, \\ a^2(4uv+2) &= \mathfrak{b} \end{aligned} \right\} \implies \Pi(z) &= (1 - 2auz + a^2z^2)(1 - 2avz + a^2z^2), \quad u, v \in \mathbb{R}. \end{aligned}$$

On a real sheet with $\Delta_{\text{fold}} \geq 0$, we additionally put

$$\xi_{\pm} = \frac{C_0 \pm \sqrt{\Delta_{\text{fold}}}}{4a}.$$

Then the pair $(u, v) = (\xi_+, \xi_-)$ satisfies the two symmetric relations in (4.9):

$$\xi_+ + \xi_- = \frac{C_0}{2a}, \quad \mathfrak{b} = a^2(4\xi_+\xi_- + 2).$$

The proof uses the root presentation only on these real sheets; the global transfer and generating identities use the polynomial Π itself. Here is a direct continuant derivation of the required folded formulas. Pair the coordinates in the order $(1, n), (2, n-1), \dots$. In this order the symmetric matrix $(xI_n - A_n)J_n - sI_n$ is block tridiagonal. Its interior diagonal and off-diagonal blocks are

$$D_{\text{int}} = \begin{bmatrix} -s & x \\ x & -s \end{bmatrix}, \quad E_{\text{off}} = \begin{bmatrix} 0 & -1 \\ -a & 0 \end{bmatrix},$$

with lower block E_{off}^T . For even order the terminal block is

$$D_{\text{term}}^e = \begin{bmatrix} -1-s & x \\ x & -a-s \end{bmatrix};$$

for odd order the terminal block is the scalar $x-s$, coupled to the preceding block by the column $(-1, -a)^T$. Since

$$(xI_n - A_n - sJ_n) = ((xI_n - A_n)J_n - sI_n)J_n,$$

the determinant of this block matrix, multiplied by $\det J_n$, is $q_n(x, s)$. Set $\mathcal{Q}_m^e = q_{2m}(x, s)$, $\mathcal{Q}_m^o = q_{2m+1}(x, s)$. We include the continuant verification to fix all boundary

signs. We now give a complete minor transfer for the folded continuant. For $\tau \in \{e, o\}$, let K_m^τ denote the corresponding matrix

$$(xI_n - A_n - sJ_n)J_n = (xI_n - A_n)J_n - sI_n, \quad n = 2m \text{ or } 2m + 1,$$

after simultaneously permuting its rows and columns into the paired order. Thus K_m^e has order $2m$, K_m^o has order $2m + 1$, and

$$K_1^e = \begin{bmatrix} -1-s & x \\ x & -a-s \end{bmatrix}, \quad K_1^o = \begin{bmatrix} -s & x & -1 \\ x & -s & -a \\ -1 & -a & x-s \end{bmatrix}.$$

For $m \geq 1$, increasing m by one adds one interior block at the exposed end:

$$(4.10) \quad K_{m+1}^\tau = \begin{bmatrix} D_{\text{int}} & \mathcal{E}_m^\tau \\ (\mathcal{E}_m^\tau)^T & K_m^\tau \end{bmatrix}, \quad \mathcal{E}_m^\tau = \begin{bmatrix} E_{\text{off}} & 0 \end{bmatrix},$$

where the zero block has the width required by K_m^τ .

For a matrix whose first two coordinates are exposed, write $K(\widehat{I} \mid \widehat{J})$ for the matrix obtained by deleting rows I and columns J . Suppressing the parity superscript, set

$$\begin{aligned} \mathfrak{d}_m &= \det K_m, & \mathfrak{p}_m &= \det K_m(\widehat{1} \mid \widehat{1}), & \mathfrak{q}_m &= \det K_m(\widehat{2} \mid \widehat{2}), \\ \mathfrak{c}_m &= \det K_m(\widehat{1} \mid \widehat{2}), & \mathfrak{r}_m &= \det K_m(\widehat{1}, \widehat{2} \mid \widehat{1}, \widehat{2}). \end{aligned}$$

The off-principal minor obtained by deleting row 2 and column 1 has the same value $\mathfrak{c}_m$, because K_m is symmetric.

Here is the one-step calculation. On the dense set on which both K_m and the tail obtained by deleting its first two coordinates are invertible, let

$$F = \begin{bmatrix} u & w \\ w & v \end{bmatrix}$$

be its Schur complement in K_m , and let δ be the determinant of that tail. Then

$$(\mathfrak{d}_m, \mathfrak{p}_m, \mathfrak{q}_m, \mathfrak{c}_m, \mathfrak{r}_m) = \delta(uv - w^2, v, u, w, 1).$$

Since

$$E_{\text{off}} \operatorname{adj}(F) E_{\text{off}}^T = \begin{bmatrix} u & -aw \\ -aw & a^2v \end{bmatrix},$$

taking the Schur complement of K_m in (4.10) gives

$$\begin{aligned} \mathfrak{d}_{m+1} &= (s^2 - x^2)\mathfrak{d}_m + s(a^2\mathfrak{p}_m + \mathfrak{q}_m) - 2ax\mathfrak{c}_m + a^2\mathfrak{r}_m, \\ \mathfrak{p}_{m+1} &= -s\mathfrak{d}_m - a^2\mathfrak{p}_m, \\ \mathfrak{q}_{m+1} &= -s\mathfrak{d}_m - \mathfrak{q}_m, \\ \mathfrak{c}_{m+1} &= x\mathfrak{d}_m + a\mathfrak{c}_m, \\ \mathfrak{r}_{m+1} &= \mathfrak{d}_m. \end{aligned} \tag{4.11}$$

Both sides are polynomial in x and s , and the nonsingular set used above is dense. The polynomial identity theorem therefore extends all five identities to the singular cases as well; no inverse is assumed in (4.11).

The five quantities in (4.11) satisfy one linear invariant. Define

$$\ell_m = (1-a)\mathfrak{d}_m - as\mathfrak{p}_m + s\mathfrak{q}_m + (1-a)x\mathfrak{c}_m - a(1-a)\mathfrak{r}_m.$$

Direct substitution into (4.11) gives

$$\ell_{m+1} = -a\ell_m.$$

For the two terminal blocks, direct evaluation gives

$$\begin{aligned} & (\mathfrak{d}_1, \mathfrak{p}_1, \mathfrak{q}_1, \mathfrak{c}_1, \mathfrak{r}_1)_e \\ &= ((1+s)(a+s) - x^2, -a-s, -1-s, x, 1), \\ & (\mathfrak{d}_1, \mathfrak{p}_1, \mathfrak{q}_1, \mathfrak{c}_1, \mathfrak{r}_1)_o \\ &= (x(1+a^2) + 2as - C_0(x-s), \\ & \quad s^2 - sx - a^2, s^2 - sx - 1, x^2 - sx - a, x - s). \end{aligned}$$

Thus $\ell_1 = 0$ in both parity classes, and consequently

$$(4.12) \quad (1-a)\mathfrak{d}_m - as\mathfrak{p}_m + s\mathfrak{q}_m + (1-a)x\mathfrak{c}_m - a(1-a)\mathfrak{r}_m = 0 \quad (m \geq 1).$$

Because

$$\det J_{2m} = \det J_{2m+1} = (-1)^m,$$

the first signed minor below is precisely the required determinant:

$$\boldsymbol{\eta}_m^\tau = (-1)^m \begin{bmatrix} \mathfrak{d}_m^\tau & \mathfrak{p}_m^\tau & \mathfrak{q}_m^\tau & \mathfrak{c}_m^\tau \end{bmatrix}^T, \quad (\boldsymbol{\eta}_m^e)_1 = \mathcal{Q}_m^e, \quad (\boldsymbol{\eta}_m^o)_1 = \mathcal{Q}_m^o.$$

Eliminating $\mathfrak{r}_m$ with (4.12) reduces (4.11) to the four-dimensional transfer

$$(4.13) \quad \boldsymbol{\eta}_{m+1}^\tau = \mathsf{T}_{\text{fold}} \boldsymbol{\eta}_m^\tau, \quad \mathsf{T}_{\text{fold}} = \begin{bmatrix} x^2 - s^2 - a & \frac{a^3 s}{1-a} & -\frac{s}{1-a} & ax \\ s & a^2 & 0 & 0 \\ s & 0 & 1 & 0 \\ -x & 0 & 0 & -a \end{bmatrix}.$$

The terminal conditions can be incorporated at $m = 0$ by setting

$$(4.14) \quad \boldsymbol{\eta}_0^e = \begin{bmatrix} 1 & a^{-1} & 1 & 0 \end{bmatrix}^T, \quad \boldsymbol{\eta}_0^o = \begin{bmatrix} x-s & 1 & 1 & -1 \end{bmatrix}^T.$$

Indeed, direct multiplication in (4.13) gives

$$\mathsf{T}_{\text{fold}} \boldsymbol{\eta}_0^e = \begin{bmatrix} x^2 - (1+s)(a+s) \\ a+s \\ 1+s \\ -x \end{bmatrix},$$

and

$$\mathsf{T}_{\text{fold}} \boldsymbol{\eta}_0^o = \begin{bmatrix} C_0(x-s) - x(1+a^2) - 2as \\ a^2 - s^2 + sx \\ 1 - s^2 + sx \\ a + sx - x^2 \end{bmatrix},$$

which are exactly the signed terminal minors above. Hence $\boldsymbol{\eta}_m^\tau = \mathsf{T}_{\text{fold}}^m \boldsymbol{\eta}_0^\tau$ for every $m \geq 0$.

A direct determinant calculation gives

$$(4.15) \quad \det(\lambda I_4 - \mathbf{T}_{\text{fold}}) = \lambda^4 - C_0 \lambda^3 + \mathfrak{b} \lambda^2 - a^2 C_0 \lambda + a^4, \\ \mathfrak{b} = (1 + a^2)x^2 + 2as^2 + 2a^2 - 2a - 2a^3.$$

Cayley–Hamilton, followed by multiplication by $\mathbf{T}_{\text{fold}}^{m-4} \boldsymbol{\eta}_0^\tau$ and extraction of the first coordinate, therefore yields, for $Z_m = \mathcal{Q}_m^e$ and for $Z_m = \mathcal{Q}_m^o$,

$$(4.16) \quad Z_m - C_0 Z_{m-1} + \mathfrak{b} Z_{m-2} - a^2 C_0 Z_{m-3} + a^4 Z_{m-4} = 0, \quad m \geq 4.$$

Finally, applying $\mathbf{T}_{\text{fold}}$ three times to (4.14) gives the boundary residuals

$$\text{Res}_0 = Z_0, \quad \text{Res}_1 = Z_1 - C_0 Z_0, \quad \text{Res}_2 = Z_2 - C_0 Z_1 + \mathfrak{b} Z_0, \\ \text{Res}_3 = Z_3 - C_0 Z_2 + \mathfrak{b} Z_1 - a^2 C_0 Z_0,$$

namely

	Res ₀	Res ₁	Res ₂	Res ₃
$Z_m = \mathcal{Q}_m^e$	1	$a(1 - 2\zeta_s)$	$a^2(1 - 2\zeta_s)$	a^3
$Z_m = \mathcal{Q}_m^o$	$x - s$	$-[x(1 + a^2) + 2as]$	$a^2(x - s)$	0

as asserted. Thus, as a formal-power-series identity,

$$\Pi(z) \sum_{m \geq 0} Z_m z^m = \text{Res}_0 + \text{Res}_1 z + \text{Res}_2 z^2 + \text{Res}_3 z^3.$$

For the even sequence the polynomial on the right factors as $(1 + az)(1 - 2a\zeta_s z + a^2 z^2)$; for the odd sequence it is $(x - s) - [x(1 + a^2) + 2as]z + a^2(x - s)z^2$. This proves (4.16). Since $\Pi(0) = 1$, it is invertible in $\mathbb{R}[[z]]$, and the two generating functions are the literal formal-power-series identities

$$(4.17) \quad \sum_{m \geq 0} \mathcal{Q}_m^e z^m = (1 + az)(1 - 2a\zeta_s z + a^2 z^2) \Pi(z)^{-1},$$

and

$$(4.18) \quad \sum_{m \geq 0} \mathcal{Q}_m^o z^m = ((x - s) - [x(1 + a^2) + 2as]z + a^2(x - s)z^2) \Pi(z)^{-1}.$$

Now let $U_{-1} = 0$. The Chebyshev generating function gives

$$\sum_{m \geq 0} (u - \zeta_s)(U_m(u) + U_{m-1}(u)) \omega_{\text{gf}}^m = \frac{(u - \zeta_s)(1 + \omega_{\text{gf}})}{1 - 2u\omega_{\text{gf}} + \omega_{\text{gf}}^2},$$

and, with $\mathbf{B}_s^{\text{fold}} = [x(1 + a^2) + 2as]/(2a)$,

$$\sum_{m \geq 0} ((x - s)u - \mathbf{B}_s^{\text{fold}}) U_m(u) \omega_{\text{gf}}^m = \frac{(x - s)u - \mathbf{B}_s^{\text{fold}}}{1 - 2u\omega_{\text{gf}} + \omega_{\text{gf}}^2}.$$

For a differentiable real function F , write

$$[F]_{u,v} := \begin{cases} \frac{F(u) - F(v)}{u - v}, & u \neq v, \\ F'(u), & u = v. \end{cases}$$

Let $u, v \in \mathbb{R}$ satisfy the two symmetric relations in (4.9). Taking the corresponding divided difference of the Chebyshev generating identities and putting $\omega_{\text{gf}} = az$ recovers (4.17) and (4.18). Coefficient comparison gives

$$(4.19) \quad q_{2m}(x, s) = a^m [\mathbf{f}_m]_{u,v}, \quad \mathbf{f}_m(t) = (t - \zeta_s)(U_m(t) + U_{m-1}(t)),$$

and

$$(4.20) \quad q_{2m+1}(x, s) = a^m [\mathbf{g}_m]_{u,v}, \quad \mathbf{g}_m(t) = \left((x - s)t - \frac{x(1 + a^2) + 2as}{2a} \right) U_m(t).$$

The determinant recurrence and generating identities are global. The real-pair formulas (4.19)–(4.20) are used only after the two symmetric relations have been verified; their collision value is the displayed derivative. No complex square-root choice is made when $\Delta_{\text{fold}} < 0$, and none is needed below.

Half-angle chord parametrization. Recall that $a = e^{-2h}$, and put

$$\mathbf{s}_h = \sinh h, \quad \mathbf{c}_h = \cosh h.$$

For $K \geq 2$ and $0 \leq \theta \leq \pi$, define

$$(4.21) \quad \Phi_K(\theta) = \sqrt{\mathbf{s}_h^2 + \sin^2 \theta} \frac{|\sin(K\theta)|}{\sin \theta},$$

where the quotient is interpreted continuously at $\theta = 0, \pi$, and

$$\Psi_K(\eta) = \sqrt{\mathbf{s}_h^2 - \sinh^2 \eta} \frac{\sinh(K\eta)}{\sinh \eta}, \quad 0 \leq \eta \leq h,$$

with the limiting value at $\eta = 0$. We spell out the half-angle reduction, including its sign. Put

$$y = \cos \theta, \quad z = \cosh \eta, \quad \mathbf{p}_\theta = \sqrt{\mathbf{c}_h^2 - y^2}, \quad \mathbf{p}_\eta = \sqrt{\mathbf{c}_h^2 - z^2},$$

and let $L_0 = 2r/\mathbf{c}_h$. If $x = L_0 yz$, $s = L_0 \mathbf{p}_\theta \mathbf{p}_\eta > 0$, and $\tau = \varsigma s$ with $\varsigma \in \{-1, 1\}$, then direct expansion gives

$$(4.22) \quad \begin{aligned} (1 + a + x)^2 - s^2 &= 4a(z + y)^2, & (1 + a - x)^2 - s^2 &= 4a(z - y)^2, \\ u = 2z^2 - 1, & v = 2y^2 - 1, & \zeta_\tau &= 2\mathbf{c}_h^2 - 1 + 2\varsigma \mathbf{p}_\theta \mathbf{p}_\eta. \end{aligned}$$

The real pair (u, v) satisfies the two symmetric relations in (4.9). For $K = 2m + 1$, use

$$U_m(2t^2 - 1) + U_{m-1}(2t^2 - 1) = U_{K-1}(t);$$

for $K = 2m + 2$, use

$$U_m(2t^2 - 1) = \frac{U_{K-1}(t)}{2t}.$$

Substitution of this pair in (4.19) and (4.20), respectively, reduces either equation $q_{K-1}(x, \tau) = 0$ to

$$(4.23) \quad \mathbf{p}_\theta U_{K-1}(y) = \varsigma \mathbf{p}_\eta U_{K-1}(z).$$

In the odd-fold case the reduction also contains the factor $\mathfrak{p}_\eta y + \varsigma \mathfrak{p}_\theta z$. On a distinct-variable sheet this factor cannot vanish: after squaring, its vanishing would give

$$(\mathfrak{c}_h^2 - z^2)y^2 = (\mathfrak{c}_h^2 - y^2)z^2,$$

and hence $y^2 = z^2$. Thus the factor is nonzero whenever the folded variables are distinct. On the outer lobe $U_{K-1}(z) > 0$, whereas, for $j\pi/K < \theta < (j+1)\pi/K$,

$$\operatorname{sgn}(U_{K-1}(\cos \theta)) = \operatorname{sgn}(\sin(K\theta)) = (-1)^j.$$

Moreover, $\mathfrak{p}_\theta |U_{K-1}(y)| = \Phi_K(\theta)$ and $\mathfrak{p}_\eta U_{K-1}(z) = \Psi_K(\eta)$. Thus (4.23) is equivalent to $\varsigma = (-1)^j$ and

$$(4.24) \quad \Phi_K(\theta) = \Psi_K(\eta),$$

with

$$(4.25) \quad s = \mathfrak{s}_{\text{hyp}}(\theta, \eta) := \frac{2r}{\mathfrak{c}_h} \sqrt{(\mathfrak{s}_h^2 + \sin^2 \theta)(\mathfrak{s}_h^2 - \sinh^2 \eta)}, \quad x = \frac{2r}{\mathfrak{c}_h} \cos \theta \cosh \eta.$$

Consequently $q_{K-1}(x, (-1)^j s) = 0$. Conversely, for distinct real folded variables, their sum and product determine

$$x^2 = L_0^2 y^2 z^2, \quad s^2 = L_0^2 (\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2).$$

Thus, for distinct real variables and a fixed sign of x , the preceding steps are reversible, and the two chord relations are equivalent to the corresponding signed equation. The determinant identities themselves extend to a folded-variable collision by the continuous divided-difference convention, but we invoke the chord equivalence only for distinct variables; the selected middle branch is proved below to be collision-free. Here $K-1$ is the matrix size. If the exterior argument has crossed through 1, write it as $z = \cos \phi$, $0 \leq \phi < \pi/K$. Then the chord equation is $\Phi_K(\theta) = \Phi_K(\phi)$, and the second factor in the formula for s is $\mathfrak{s}_h^2 + \sin^2 \phi$. We denote the resulting value by

$$\mathfrak{s}_{\text{ell}}(\theta, \phi) = \frac{2r}{\mathfrak{c}_h} \sqrt{(\mathfrak{s}_h^2 + \sin^2 \theta)(\mathfrak{s}_h^2 + \sin^2 \phi)}.$$

Branch classification and exhaustion. We next identify exactly which of these chords is the middle branch of Lemma 4.2. Set

$$(4.26) \quad \mathcal{W}_K(u) = \sqrt{\mathfrak{c}_h^2 - u^2} |U_{K-1}(u)|, \quad |u| \leq \mathfrak{c}_h.$$

At $u = \pm \mathfrak{c}_h$, this denotes the continuous endpoint value 0. Then

$$\Phi_K(\theta) = \mathcal{W}_K(\cos \theta), \quad \Psi_K(\eta) = \mathcal{W}_K(\cosh \eta).$$

If $u_{\ell, K}^{(0)} = \cos \frac{\ell\pi}{K}$, then, away from the zeros,

$$\frac{d}{du} \log \mathcal{W}_K(u) = \sum_{\ell=1}^{K-1} \frac{1}{u - u_{\ell, K}^{(0)}} - \frac{u}{\mathfrak{c}_h^2 - u^2},$$

and its derivative is

$$(4.27) \quad - \sum_{\ell=1}^{K-1} \frac{1}{(u - u_{\ell, K}^{(0)})^2} - \frac{\mathfrak{c}_h^2 + u^2}{(\mathfrak{c}_h^2 - u^2)^2} < 0.$$

At each finite nodal endpoint the logarithmic derivative tends to the appropriate signed infinity, and it tends to $-\infty$ as $u \uparrow \mathfrak{c}_h$. Thus there is exactly one lobe maximum in every nodal interval, including the outer interval $\mathcal{J}_K^{\text{out}} = (\cos \frac{\pi}{K}, \mathfrak{c}_h)$. Let u_K^{max} be the unique maximizer on $\mathcal{J}_K^{\text{out}}$. For every $0 \leq \ell < \mathcal{W}_K(u_K^{\text{max}})$, there is a unique solution

$$z_K(\ell) \in (u_K^{\text{max}}, \mathfrak{c}_h], \quad \mathcal{W}_K(z_K(\ell)) = \ell;$$

we call it the *endpoint-side outer solution*. If $z_K(\ell) \geq 1$, write it as $\cosh \eta$, $0 \leq \eta \leq h$; if it is at most 1, write it as $\cos \phi$, $0 \leq \phi < \pi/K$. These are two coordinates on the same outer lobe, joined at $z = 1$.

For a positive-half gap and $\theta \in (j\pi/K, (j+1)\pi/K)$, $\theta \leq \pi/2$, a *j-chord* is the pair

$$y = \cos \theta, \quad z = z_K(\mathcal{W}_K(y)), \quad \mathcal{W}_K(y) = \mathcal{W}_K(z),$$

together with the coordinate-free values

$$x = L_0 y z, \quad s = L_0 \sqrt{(\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2)};$$

its signed root is $(-1)^j s$. The term *endpoint-side* always refers to the part of the outer lobe adjacent to $\mathfrak{c}_h$, not to the hyperbolic or elliptic coordinate used for it.

We now prove that these chords are exactly the middle signed branch. By reflection it suffices to consider a positive-half gap, so $j \geq 1$ and $\theta \leq \pi/2$. Put $\phi_{\text{rem}} = \theta - j\pi/K \in (0, \pi/K)$. Since $|\sin(K\phi_{\text{rem}})| = |\sin(K\theta)|$ and

$$\mathcal{F}_h(t) = \frac{\sqrt{\mathfrak{s}_h^2 + \sin^2 t}}{\sin t}$$

is strictly decreasing on $(0, \pi/2]$,

$$(4.28) \quad \Phi_K(\theta) < \Phi_K(\phi_{\text{rem}}) \leq \max_{u \in \mathcal{J}_K^{\text{out}}} \mathcal{W}_K(u).$$

Thus the endpoint-side outer solution exists for every inner parameter.

At $x = \lambda_{K-1,k} = 2r \cos \frac{k\pi}{K}$, $s = 0$, the two folded variables are $\cos \frac{2k\pi}{K}$ and $\cosh(2h)$, or, in half variables, $\cos \frac{k\pi}{K}$ and $\mathfrak{c}_h$. Moreover, zero is a simple eigenvalue of $B_{K-1}(\lambda_{K-1,k})$: the matrix $\lambda_{K-1,k} I_{K-1} - A_{K-1}$ has rank $K-2$, and J_{K-1} is invertible. The implicit-function theorem therefore gives, near either gap endpoint, a real continuation of the middle signed root and real half variables $y = \cos \theta$ and z .

Continue this ordered middle root $s_*(x)$ through the open gap. Write $\Delta_*(x) = \Delta_{\text{fold}}(x, s_*(x))$ for its folded discriminant, let $y(x) \leq z(x)$ be the reconstructed non-negative half variables on the real sheet issuing from the endpoint germ, and set $\Theta(x) = \arccos y(x)$. Let $\mathcal{C}$ be the set of gap points for which

$$\Delta_*(x) > 0, \quad \frac{j\pi}{K} < \Theta(x) < \frac{(j+1)\pi}{K}, \quad u_K^{\text{max}} < z(x) < \mathfrak{c}_h.$$

On $\mathcal{C}$, the reversible folded determinant identities give $\mathcal{W}_K(y(x)) = \mathcal{W}_K(z(x))$; uniqueness on the endpoint-side outer lobe then gives

$$z(x) = z_K(\mathcal{W}_K(y(x))).$$

Thus $\mathcal{C}$ is precisely the selected endpoint-side chord sheet. The endpoint germ just constructed makes $\mathcal{C}$ nonempty. The ordered eigenvalues of the continuous real

symmetric pencil $B_{K-1}(x)$ are continuous in x , including at an eigenvalue crossing, so a multiple least singular value does not interrupt the continuation. The folded pair $u = 2z^2 - 1$, $v = 2y^2 - 1$ and its reconstructed half variables are continuous, while the sheet inequalities are strict. Thus $\mathcal{C}$ is relatively open in the gap.

It is also relatively closed. At a relative closure point the chord identity and all sheet inequalities persist weakly. If the weak limit had $\Delta_*(x) = 0$, then its two reconstructed half variables would coincide. This is impossible because

$$z \geq u_K^{\max} > \cos \frac{\pi}{K} \geq \cos \frac{j\pi}{K} \geq y.$$

The signs of the half variables cannot change in a noncentral positive gap: $z > 0$ throughout, while $y = 0$ would force $x = 0$, outside such a gap. Moreover, z cannot reach $\mathfrak{c}_h$ in the interior, because (4.25) would then give $s = 0$. The inner variable cannot reach a nodal endpoint: the common level would be zero, the endpoint-side solution would be $\mathfrak{c}_h$, and (4.25) would give $s = 0$, whereas the middle root has no zero inside a spectral gap. The outer variable cannot leave its endpoint-side component, since its first such point would be $u_K^{\max}$, contradicting (4.28). Passage through $z = 1$ is harmless; only its coordinate changes. Hence every relative closure point remains in $\mathcal{C}$. Since the open gap is an interval, $\mathcal{C}$ is the entire gap. This proves the first exhaustion direction: every actual gap point lies on the selected endpoint-side chord sheet.

For the converse direction, the reconstructed angle Θ extends continuously to the closed spectral gap. The exact folded variables at the two endpoints give

$$\Theta(\lambda_{K-1,j+1}) = \frac{(j+1)\pi}{K}, \quad \Theta(\lambda_{K-1,j}) = \frac{j\pi}{K}.$$

For every $\theta_0 \in (j\pi/K, (j+1)\pi/K)$, the intermediate value theorem therefore supplies an interior gap point x with $\Theta(x) = \theta_0$. The classification just proved and uniqueness of the endpoint-side outer solution then give

$$y(x) = \cos \theta_0, \quad z(x) = z_K(\mathcal{W}_K(\cos \theta_0)),$$

so the corresponding abscissa and signed root are exactly those of the prescribed j -chord. Thus every chord parameter produces an actual point of the ordered middle branch. By Lemma 4.2, its positive magnitude is $s_{\min}(xI_{K-1} - A_{K-1})$. Both exhaustion directions have now been proved without assuming that $\theta \mapsto x$ is monotone.

When K is odd, the same argument applies to the positive half of the single central gap, continuing from the positive spectral endpoint to $x = 0$. Because $z > 0$, the identity $x = (2r/\mathfrak{c}_h) \cos \theta z$ forces $\theta = \pi/2$ at $x = 0$. Thus every parameter on that half occurs, and reflection covers the negative half.

Noncentral gap comparison. We now compare the noncentral middle branches. Let $N \geq 3$, put $M = N + 1$, and set $\kappa = N/M$. Every positive noncentral gap of the M -problem satisfies

$$(4.29) \quad j \geq 1, \quad \frac{(j+1)\pi}{N} \leq \frac{\pi}{2}.$$

Indeed, such gaps satisfy $j+1 \leq N/2$. Hence $\theta_0 = \theta/\kappa$ belongs to the same-indexed N -nodal interval, and every trigonometric argument below lies in $(0, \pi/2]$. Consider a point of such a branch for the matrix of order $M - 1$, with

$$\theta \in \left(\frac{j\pi}{M}, \frac{(j+1)\pi}{M} \right), \quad \frac{(j+1)\pi}{N} \leq \frac{\pi}{2}.$$

First suppose $\Phi_M(\theta) = \Psi_M(\eta)$. Put $\theta_0 = \theta/\kappa$. Then θ_0 belongs to the corresponding N -nodal interval. Concavity of sine gives

$$(4.30) \quad \frac{\Phi_N(\theta_0)}{\Phi_M(\theta)} = \frac{\mathcal{F}_h(\theta_0)}{\mathcal{F}_h(\theta)} > \kappa.$$

Indeed, $\theta = \kappa\theta_0$ and strict concavity gives $\sin \theta > \kappa \sin \theta_0$; hence

$$\mathcal{F}_h(\theta_0)^2 - \kappa^2 \mathcal{F}_h(\theta)^2 = (1 - \kappa^2) + \mathfrak{s}_h^2 \left(\frac{1}{\sin^2 \theta_0} - \frac{\kappa^2}{\sin^2 \theta} \right) > 0.$$

On the other hand,

$$(4.31) \quad \frac{\Psi_N(\eta)}{\Psi_M(\eta)} = \frac{\sinh(N\eta)}{\sinh(M\eta)} \leq \kappa,$$

with strict inequality for $\eta > 0$, by convexity of $\sinh$ on the positive half-axis. Hence $\Phi_N(\theta_0) > \Psi_N(\eta)$. Since Φ_N vanishes at the right endpoint $(j+1)\pi/N$, its rightmost solution $\widehat{\theta}$ of

$$\Phi_N(\widehat{\theta}) = \Psi_N(\eta)$$

exists by the intermediate value theorem and satisfies $\widehat{\theta} > \theta_0 > \theta$. Strict log-concavity of $\mathcal{W}_N$ makes it the unique solution on the descending half of that lobe. The same exterior variable is endpoint-side for the N -problem as well. Indeed, for $u = \cosh \eta > 1$,

$$\frac{d}{du} \log \mathcal{W}_M(u) - \frac{d}{du} \log \mathcal{W}_N(u) = \frac{M \coth(M\eta) - N \coth(N\eta)}{\sinh \eta} > 0.$$

The numerator is positive because, for $\eta > 0$,

$$\frac{d}{dt} (t \coth(t\eta)) = \frac{\sinh(t\eta) \cosh(t\eta) - t\eta}{\sinh^2(t\eta)} > 0.$$

The endpoint-side branch is characterized by a nonpositive logarithmic derivative. The displayed difference therefore makes the N -derivative strictly negative. The case $\eta = 0$ follows by continuity. The chord classification above says that the constructed value $\mathfrak{s}_{\text{hyp}}(\widehat{\theta}, \eta)$ is the least singular value at its constructed abscissa. Formula (4.25) and the strict increase of sine on the interval now give

$$(4.32) \quad \mathfrak{s}_{\text{hyp}}(\widehat{\theta}, \eta) > \mathfrak{s}_{\text{hyp}}(\theta, \eta).$$

Suppose next that the exterior endpoint is elliptic, so that $\Phi_M(\theta) = \Phi_M(\phi)$, $0 < \phi < \pi/M$. (The case $\phi = 0$ is the limiting hyperbolic case already covered.) Put

$$\theta_0 = \theta/\kappa, \quad \phi_0 = \phi/\kappa.$$

The ratio

$$\mathcal{A}_{N,M}^\Phi(t) = \frac{\Phi_N(t/\kappa)}{\Phi_M(t)} = \frac{\mathcal{F}_h(t/\kappa)}{\mathcal{F}_h(t)}$$

is strictly increasing while $0 < t < t/\kappa \leq \pi/2$. Indeed,

$$\frac{d}{dt} \log \mathcal{A}_{N,M}^\Phi(t) = \mathfrak{d}_h(t) - \kappa^{-1} \mathfrak{d}_h(t/\kappa) > 0, \quad \mathfrak{d}_h(t) = \frac{\mathfrak{s}_h^2 \cot t}{\mathfrak{s}_h^2 + \sin^2 t},$$

because

$$t\mathfrak{d}_h(t) = (t \cot t) \frac{\mathfrak{s}_h^2}{\mathfrak{s}_h^2 + \sin^2 t}$$

is strictly decreasing: both positive factors decrease on $(0, \pi/2)$, and the second decreases strictly. Since $\theta > \pi/M > \phi$, it follows that $\Phi_N(\theta_0) > \Phi_N(\phi_0)$. The rightmost solution of

$$\Phi_N(\widehat{\theta}) = \Phi_N(\phi_0)$$

therefore has $\widehat{\theta} > \theta_0$. Its exterior variable is endpoint-side. Indeed, if $\Lambda_K(t) = (\log \Phi_K)'(t)$, then on the endpoint-side part of the first outer lobe $\Lambda_M(\phi) \geq 0$, and differentiation of $\Phi_N(t/\kappa) = \mathcal{A}_{N,M}^\Phi(t)\Phi_M(t)$ gives

$$\kappa^{-1}\Lambda_N(t/\kappa) = \Lambda_M(t) + \frac{(\mathcal{A}_{N,M}^\Phi)'(t)}{\mathcal{A}_{N,M}^\Phi(t)} > 0.$$

Because $\Lambda_K(\phi) = -\sin \phi (\log \mathcal{W}_K)'(\cos \phi)$, positivity of $\Lambda_N(\phi_0)$ is exactly the endpoint-side condition for the N -outer solution. The one-maximum property now fixes the label, and the chord classification says again that $\mathfrak{s}_{\text{ell}}(\widehat{\theta}, \phi_0)$ is the least singular value at the constructed abscissa. Here $\widehat{\theta} > \theta_0 > \theta$ and $\phi_0 > \phi$, with all four angles below $\pi/2$. Thus both sine factors in the elliptic version of (4.25) have increased, at least one strictly, and hence

$$(4.33) \quad \mathfrak{s}_{\text{ell}}(\widehat{\theta}, \phi_0) > \mathfrak{s}_{\text{ell}}(\theta, \phi).$$

Central gaps. It remains to compare the central sheets. We do this in the next calculation.

For $m \geq 1$, let

$$(4.34) \quad c_m = s_{\min}(aI_m + V_m).$$

By reflection, the two gaps adjacent to the zero eigenvalue of A_{2m+1} have the same height; denote it by

$$(4.35) \quad d_m = \max_{0 \leq x \leq 2r \sin \frac{\pi}{2m+2}} s_{\min}(xI_{2m+1} - A_{2m+1}).$$

We claim

$$(4.36) \quad c_m > d_m > c_{m+1}.$$

Put

$$\zeta = \frac{a + a^{-1}}{2}.$$

For every $j \geq 1$, there is a unique $\rho_j > \cos \frac{\pi}{j+1}$ such that

$$(4.37) \quad aU_j(\rho_j) = U_{j-1}(\rho_j).$$

Indeed, the Christoffel–Darboux identity gives, above the largest zero of U_j ,

$$\frac{d}{dt} \frac{U_{j-1}(t)}{U_j(t)} = -\frac{2 \sum_{k=0}^{j-1} U_k(t)^2}{U_j(t)^2} < 0.$$

573 The ratio decreases from $+\infty$ to 0, proving existence and uniqueness. A three-term
 574 determinant recurrence gives

$$575 \quad (4.38) \quad \det((aI_j + V_j)^*(aI_j + V_j) - (1 + a^2 - 2at)I_j) = a^j (U_j(t) - a^{-1}U_{j-1}(t)).$$

576 Thus the largest root ρ_j gives the least eigenvalue. Together with

$$577 \quad U_j(t)^2 + U_{j-1}(t)^2 - 2tU_j(t)U_{j-1}(t) = 1,$$

578 equations (4.37) and (4.38) give

$$579 \quad (4.39) \quad c_j^2 = 1 + a^2 - 2a\rho_j = \frac{1}{U_j(\rho_j)^2}, \quad \zeta - \rho_j = \frac{1}{2U_{j-1}(\rho_j)U_j(\rho_j)}.$$

580 We shall also use that c_m is exactly the height of the central gap of A_{2m} . A
 581 permutation separating odd and even coordinates gives

$$582 \quad A_{2m} \sim \begin{bmatrix} 0 & I_m + aV_m^T \\ aI_m + V_m & 0 \end{bmatrix}.$$

583 Moreover,

$$584 \quad (I_m + aV_m^T)^*(I_m + aV_m^T) - J_m(aI_m + V_m)^*(aI_m + V_m)J_m = (1 - a^2)\mathbf{e}_m\mathbf{e}_m^T \geq 0.$$

585 The singular values of the block off-diagonal matrix are the union of the singular values
 586 of its two off-diagonal blocks. The Gram inequality shows that $s_{\min}(I_m + aV_m^T) \geq$
 587 $s_{\min}(aI_m + V_m) = c_m$, and hence $s_{\min}(A_{2m}) = c_m$. To prove that this is the maximum
 588 throughout the central gap, put $K = 2m + 1$ and use the half variables

$$589 \quad y = \cos \theta, \quad z = \cosh \eta \quad \text{or} \quad z = \cos \phi.$$

590 On the positive half of the central middle branch, the preceding chord classification
 591 gives

$$592 \quad (4.40) \quad \mathcal{W}_K(y) = \mathcal{W}_K(z), \quad x = \frac{2r}{\mathbf{c}_h}yz, \quad s^2 = \frac{4a}{\mathbf{c}_h^2}(\mathbf{c}_h^2 - y^2)(\mathbf{c}_h^2 - z^2),$$

593 where $0 \leq y < \cos \frac{m\pi}{K}$ is in the central lobe and z is the endpoint-side outer solution.

594 The function $\mathcal{W}_K$ is even, and strict log-concavity on the central lobe gives

$$595 \quad \mathcal{W}_K(y) < \mathcal{W}_K(0) = \mathbf{c}_h \quad (y > 0).$$

596 Let

$$597 \quad z_0 = \sqrt{\frac{1 + \rho_m}{2}}.$$

598 The half-angle identity and (4.37) give

$$599 \quad U_{2m}(z_0) = U_m(\rho_m) + U_{m-1}(\rho_m) = (1 + a)U_m(\rho_m).$$

600 Together with (4.39), this yields

$$601 \quad (4.41) \quad \mathcal{W}_K(z_0) = \mathbf{c}_h, \quad 4a(\mathbf{c}_h^2 - z_0^2) = 2a(\zeta - \rho_m) = c_m^2.$$

602 At $x = 0$, Lemma 4.2 and the chord-exhaustion argument identify the nearest root
 603 with c_m , so z_0 is the endpoint-side outer solution of level $\mathbf{c}_h$. If $x \neq 0$, the level in (4.40)

is strictly below $\mathfrak{c}_h$, and hence its endpoint-side solution satisfies $z > z_0$. Equations (4.40)–(4.41) then give

$$s^2 < 4a(\mathfrak{c}_h^2 - z_0^2) = c_m^2.$$

Reflection handles the negative half of the gap. Thus its height is exactly c_m , attained only at $x = 0$.

We first prove $d_m < c_m$. Set

$$\omega_m^e = \cos \frac{\pi}{m+1}, \quad \mathcal{V}_m(t) = \sqrt{(\zeta - t)(1 + t)} U_m(t).$$

Here are the details of the central specialization. Take $K = 2m + 2$ and set

$$t_{\text{in}} = 2y^2 - 1 = \cos(2\theta), \quad t_{\text{out}} = 2z^2 - 1.$$

On the positive central gap, $-1 < t_{\text{in}} < -\omega_m^e$, while the endpoint-side outer variable satisfies $\omega_m^e < t_{\text{out}} < \zeta$, where $\zeta = 2\mathfrak{c}_h^2 - 1$. The identity $U_{2m+1}(t) = 2tU_m(2t^2 - 1)$ gives

$$\mathcal{W}_{2m+2}(y)^2 = (\zeta - t_{\text{in}})(1 + t_{\text{in}})U_m(t_{\text{in}})^2, \quad \mathcal{W}_{2m+2}(z)^2 = (\zeta - t_{\text{out}})(1 + t_{\text{out}})U_m(t_{\text{out}})^2.$$

Because $\zeta + 1 = 2\mathfrak{c}_h^2$, the formulas $x = (2r/\mathfrak{c}_h)yz$ and $s^2 = (4a/\mathfrak{c}_h^2)(\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2)$ become the following chord parametrization:

$$(4.42) \quad \begin{aligned} -1 < t_{\text{in}} < -\omega_m^e, \quad \omega_m^e < t_{\text{out}} < \zeta, \quad \mathcal{V}_m(t_{\text{out}}) = |\mathcal{V}_m(t_{\text{in}})|, \\ x^2 = \frac{2a(1 + t_{\text{out}})(1 + t_{\text{in}})}{\zeta + 1}, \quad s^2 = \frac{2a(\zeta - t_{\text{out}})(\zeta - t_{\text{in}})}{\zeta + 1}, \end{aligned}$$

where t_{out} is the descending, or largest, solution. To see this from (4.24), use

$$\mathfrak{s}_h^2 + \sin^2 \theta = \frac{\zeta - t_{\text{in}}}{2}, \quad \mathfrak{s}_h^2 - \sinh^2 \eta = \frac{\zeta - t_{\text{out}}}{2}.$$

Continuation through $t_{\text{out}} = 1$ uses the elliptic coordinate $z = \cos \phi$.

Let $\rho = \rho_m$. The function $\log \mathcal{V}_m$ is strictly concave on (ω_m^e, ζ) . If $(\log \mathcal{V}_m)'(\rho) \geq 0$, the descending root satisfies $t_{\text{out}} \geq \rho$, because ρ lies at or before the unique lobe maximum and the descending root lies at or after it. Suppose instead that $(\log \mathcal{V}_m)'(\rho) < 0$. Then $\rho > a$. Indeed, otherwise $\rho \leq a < 1$, so $\rho < 1$ and in particular $\rho^2 - 1 \neq 0$. The identity

$$\frac{U'_m(\rho)}{U_m(\rho)} = \frac{m\rho - (m+1)a}{\rho^2 - 1}$$

would give

$$\begin{aligned} (\log \mathcal{V}_m)'(\rho) &= \frac{(2m+2)a + 1 - (2m+1)\rho}{2(1 - \rho^2)} - \frac{1}{2(\zeta - \rho)} \\ &\geq \frac{1}{2(1 - \rho)} - \frac{1}{2(\zeta - \rho)} \geq 0, \end{aligned}$$

a contradiction. Thus the displayed quotient has been used only on its valid domain; if $\rho = 1$, the desired conclusion $\rho > a$ is immediate from $a < 1$ and no quotient is invoked. Writing $t_{\text{in}} = -\cos \phi_c$, $0 < \phi_c < \pi/(m+1)$, gives

$$\begin{aligned} |\mathcal{V}_m(t_{\text{in}})|^2 &= \frac{(\zeta + \cos \phi_c) \sin^2((m+1)\phi_c)}{1 + \cos \phi_c} \\ &< \frac{\zeta + \omega_m^e}{1 + \omega_m^e} < \frac{1 + a}{2a} < \frac{1 + \rho}{2a} = \mathcal{V}_m(\rho)^2. \end{aligned}$$

Now ρ is on the descending side and $|\mathcal{V}_m(t_{\text{in}})| < \mathcal{V}_m(\rho)$, so the unique descending root again satisfies $t_{\text{out}} > \rho$. In either case, $t_{\text{out}} \geq \rho$, and $t_{\text{in}} > -1$ makes the following inequality strict:

$$s^2 < 2a(\zeta - \rho) = c_m^2.$$

This holds along the whole central sheet, so $d_m < c_m$.

We now prove $d_m > c_{m+1}$. Put $L = m + 1 \geq 2$, and abbreviate

$$\rho = \rho_L, \quad u_- = U_{L-1}(\rho), \quad u_+ = U_L(\rho), \quad \omega_L = \cos \frac{\pi}{2L}, \quad \Delta_\omega = 1 - \omega_L,$$

$$(4.43) \quad \mathfrak{D}_L = 1 + 2(\rho + \omega_L)u_-u_+, \quad \xi = \rho - \frac{\Delta_\omega}{\mathfrak{D}_L}.$$

Thus $a = u_-/u_+$ and $c_L = 1/u_+$. We first establish the scalar inequality

$$(4.44) \quad (1 + \omega_L)(1 + \xi)^2 U_{L-1}(\xi)^2 < (\rho + \omega_L)\mathfrak{D}_L.$$

Put $b = u_+/u_- = 1/a$, and define

$$Q_b = b^2 - 2b\rho + 1, \quad S_b = b^2 + 2b\omega_L + 1.$$

The Chebyshev identity gives $u_-^2 Q_b = 1$, and

$$(4.45) \quad \mathfrak{D}_L = S_b/Q_b, \quad \Delta_\xi := \rho - \xi = \Delta_\omega Q_b/S_b.$$

Indeed, $u_+ = bu_-$ and $u_-^2 = Q_b^{-1}$, so

$$\mathfrak{D}_L = 1 + 2(\rho + \omega_L)u_-u_+ = 1 + \frac{2b(\rho + \omega_L)}{Q_b} = \frac{S_b}{Q_b},$$

and the formula for Δ_ξ follows from (4.43). The ratio U_L/U_{L-1} is strictly increasing above the largest zero of U_{L-1} , and equals 1 at $\rho_{\text{ref}} = \cos \frac{\pi}{2L+1}$. Hence $\rho > \rho_{\text{ref}}$. Since $S_b - Q_b = 2b(\rho + \omega_L) > 0$,

$$\xi > \rho_{\text{ref}} - \Delta_\omega > 2\omega_L - 1 > 2\omega_L^2 - 1 = \cos \frac{\pi}{L}.$$

Therefore $f_L(x) = (1 + x)U_{L-1}(x)$ is positive, strictly increasing, and strictly log-concave on $[\xi, \rho]$. In particular, $\mathfrak{d}_\rho = (\log f_L)'(\rho) > 0$, and

$$\mathfrak{d}_\rho = \begin{cases} \frac{L(b - \rho) - 1}{\rho^2 - 1}, & \rho \neq 1, \\ \frac{2L^2 + 1}{6}, & \rho = 1. \end{cases}$$

The endpoint value follows directly from $U_{L-1}(1) = L$ and $U'_{L-1}(1) = L(L^2 - 1)/3$. Moreover,

$$\log f_L(\rho) - \log f_L(\xi) > \Delta_\xi \mathfrak{d}_\rho, \quad f_L(\xi)^2 < f_L(\rho)^2 e^{-2\Delta_\xi \mathfrak{d}_\rho} < \frac{f_L(\rho)^2}{1 + 2\Delta_\xi \mathfrak{d}_\rho}.$$

Here the first strict inequality is the integral form of strict log-concavity. Since $f_L(\rho)^2 = (1 + \rho)^2/Q_b$ and $\Delta_\xi = \Delta_\omega Q_b/S_b$, substitution into (4.44) reduces it to

$$(4.46) \quad \mathcal{N} = (\rho + \omega_L)(S_b + 2\Delta_\omega Q_b \mathfrak{d}_\rho) - (1 + \omega_L)(1 + \rho)^2 > 0.$$

We shall repeatedly use

$$(4.47) \quad \Delta_\omega = 1 - \cos \frac{\pi}{2L} < \frac{\pi^2}{8L^2} < \frac{5}{4L^2}, \quad \Delta_\omega L < 1.$$

Suppose first that $\rho > 1$. Put $u = b - \rho$ and $\Delta_+ = \rho^2 - 1$. Since $\mathfrak{d}_\rho > 0$, $u > 1/L$. Keeping ρ fixed, direct differentiation gives

$$\frac{\partial_b \mathcal{N}}{2(\rho + \omega_L)} = \rho + \omega_L + u - \Delta_\omega L + \frac{\Delta_\omega(3Lu^2 - 2u)}{\Delta_+}.$$

This is strictly increasing in $u \geq 1/L$, and its value at $1/L$ is positive because $\Delta_\omega L < 1$. Thus $\mathcal{N}$ exceeds its value at $b = \rho + 1/L$. With $s_0 = \rho + \omega_L$ and $\omega_+ = 1 + \omega_L$, that value is

$$\Upsilon(s_0) = s_0 \left[\left(s_0 + \frac{1}{L} \right)^2 + \omega_+ \Delta_\omega \right] - \omega_+ (s_0 + \Delta_\omega)^2.$$

For $s_0 \geq \omega_+$,

$$\Upsilon''(s_0) = 6s_0 + 4/L - 2\omega_+ > 0, \quad \Upsilon'(\omega_+) = 2\omega_L \omega_+ + 4\omega_+/L + 1/L^2 > 0,$$

and

$$\Upsilon(\omega_+) = \omega_+ (-2\Delta_\omega + 2\omega_+/L + 1/L^2) > 0.$$

Thus (4.46) holds when $\rho > 1$. If $\rho = 1$, then $b = (L + 1)/L$, $Q_b = L^{-2}$, and direct substitution, using the endpoint value of $\mathfrak{d}_\rho$, gives

$$\mathcal{N} = \Upsilon(1 + \omega_L) + \frac{2(1 + \omega_L)\Delta_\omega}{L^2} \frac{2L^2 + 1}{6} > 0.$$

The first term is positive by the preceding calculation, and every factor in the correction term is positive.

Suppose next that $\rho < 1$. Set

$$v = 1 - \rho, \quad \Delta_b = b - 1, \quad u = \Delta_b + v, \quad \Delta_- = v(2 - v).$$

Writing $\rho = \cos \theta_\rho$ gives $0 < L\theta_\rho < \pi/2$ and

$$u = b - \rho = \frac{\sin((L + 1)\theta_\rho)}{\sin(L\theta_\rho)} - \cos \theta_\rho = \sin \theta_\rho \cot(L\theta_\rho).$$

Because $b > 1$, $u = \Delta_b + v > v > 0$. Also $\tan(L\theta_\rho) > L \tan \theta_\rho > L \sin \theta_\rho$, since $x \mapsto \tan x/x$ is strictly increasing on $(0, \pi/2)$. Hence

$$(4.48) \quad 0 < v < u < 1/L.$$

With ρ fixed, write $\mathcal{N}_\rho^<(\cdot)$ for $\mathcal{N}$ viewed as a function of $\Delta_b = b - 1$. A direct expansion of (4.46) gives

$$\mathcal{N}_\rho^<(\Delta_b) - \mathcal{N}_\rho^<(0) = \frac{\Delta_b(\rho + \omega_L)}{\Delta_-} \mathfrak{q}(u),$$

where

$$\mathfrak{q}(u) = \Delta_- [2(2 - \Delta_\omega) + u - v] + 2\Delta_\omega [u + v - L(u^2 + uv + v^2) - L\Delta_-].$$

695 This is a concave quadratic. Moreover,

$$696 \quad \mathcal{N}_\rho^<(0) = \frac{v}{2-v} [(2 - \Delta_\omega)(2 - v)^2 - 2\Delta_\omega^2 - 4\Delta_\omega L(2 - v - \Delta_\omega)].$$

697 The elementary bounds, the second following from $\rho > \rho_{\text{ref}}$ and $1 - \rho_{\text{ref}} < \pi^2/[2(2L +$
 698 $1)^2] < 5/(2L + 1)^2$,

$$699 \quad (4.49) \quad \Delta_\omega < \frac{5}{4L^2}, \quad v < \frac{5}{(2L + 1)^2} \leq \frac{1}{5}.$$

700 The last inequality is an equality when $L = 2$; because the preceding bound for v
 701 is strict, one still has $v < 1/5$. These bounds show that the bracketed expression is
 702 positive, since it equals

$$703 \quad (2 - v)[(2 - \Delta_\omega)(2 - v) - 4\Delta_\omega L] + 2\Delta_\omega^2(2L - 1)$$

704 and, using $L \geq 2$,

$$705 \quad (2 - \Delta_\omega)(2 - v) > \frac{27}{16} \frac{9}{5} = \frac{243}{80} > \frac{5}{2} \geq 4\Delta_\omega L.$$

706 At the endpoints of (4.48),

$$707 \quad \mathbf{q}(v) = v[(2 - v)(4 - 2\Delta_\omega) + 4\Delta_\omega - 4\Delta_\omega L(1 + v)] > 0,$$

708

$$709 \quad \mathbf{q}(1/L) = v[(2 - v)(4 - 2\Delta_\omega + 1/L - v) - 4\Delta_\omega L] > 0.$$

710 Indeed, in these two lines the positive leading terms exceed

$$711 \quad \frac{9}{5} \left(4 - \frac{5}{8}\right) = \frac{243}{40}, \quad \frac{9}{5} \left(4 - \frac{5}{8} - \frac{1}{5}\right) = \frac{1143}{200},$$

712 while the negative terms have magnitudes less than $(5/L)(6/5) \leq 3$ and $5/L \leq 5/2$,
 713 respectively. Concavity now gives $\mathbf{q}(u) > 0$, proving (4.46), and hence (4.44), in all
 714 cases.

715 It remains to certify that the scalar comparison concerns the least singular value.
 716 Since $a > 0$, $\Delta_\omega > 0$, $\rho + \omega_L > 0$, and $\zeta + \omega_L > 0$, let $x_* > 0$ be the positive square
 717 root determined by

$$718 \quad (4.50) \quad x_*^2 = \frac{2a\Delta_\omega(\rho + \omega_L)}{\zeta + \omega_L}, \quad c = c_L.$$

719 At $x = x_*$ and $s = \pm c$, the real pair $u = \xi$, $v = -\omega_L$ satisfies the two symmetric
 720 relations in (4.9). Indeed, its sum and product are $\xi - \omega_L$ and $-\omega_L \xi$, and the required
 721 identities follow from

$$722 \quad \mathfrak{D}_L = \frac{\zeta + \omega_L}{\zeta - \rho}, \quad \xi = \frac{\rho(\zeta + 1) + \zeta(\omega_L - 1)}{\zeta + \omega_L}.$$

723 Moreover, $\xi + \omega_L > 0$, so the distinct branch of the divided difference in (4.20) applies.
 724 For convenience, write the folded numerator as

$$725 \quad \mathbf{n}_s(t) = -[x(\zeta - t) + s(1 + t)]U_{L-1}(t).$$

Put

$$X_\xi = x_*(\zeta - \xi), \quad Y_\xi = c(1 + \xi), \quad X_\omega = x_*(\zeta + \omega_L), \quad Y_\omega = c\Delta_\omega,$$

and put $U_\xi = U_{L-1}(\xi)$, $U_{-\omega} = U_{L-1}(-\omega_L)$. The identities

$$\mathfrak{D}_L = 2a(\zeta + \omega_L)u_+^2, \quad \mathfrak{D}_L(\rho + \omega_L)(\zeta - \xi) = (\zeta + \omega_L)(1 + \xi)$$

give $Y_\omega Y_\xi = X_\omega X_\xi$. The identity $u_-^2 Q_b = 1$ gives $Q_b > 0$, and $S_b - Q_b = 2b(\rho + \omega_L) > 0$, so $\mathfrak{D}_L = S_b/Q_b > 1$. Moreover, $L \geq 2$ gives $\omega_L > 1/2$, while $\rho > \rho_{\text{ref}} > \omega_L$; hence

$$\rho + \omega_L > 2\omega_L > 1 > 1 - \omega_L > 0.$$

Consequently,

$$\frac{X_\omega^2}{Y_\omega^2} = \frac{\mathfrak{D}_L(\rho + \omega_L)}{1 - \omega_L} > 1,$$

so, with $\mathfrak{r}_* = X_\xi/Y_\xi = Y_\omega/X_\omega \in (0, 1)$, the common nonzero factor in (4.20) is $a^{L-1}/(\xi + \omega_L)$, and the two numerator differences are

$$\begin{aligned} \mathfrak{n}_c(\xi) - \mathfrak{n}_c(-\omega_L) &= (1 + \mathfrak{r}_*)(X_\omega U_{-\omega} - Y_\xi U_\xi), \\ \mathfrak{n}_{-c}(\xi) - \mathfrak{n}_{-c}(-\omega_L) &= (1 - \mathfrak{r}_*)(X_\omega U_{-\omega} + Y_\xi U_\xi). \end{aligned}$$

Their product is positive by (4.44), because

$$U_{-\omega}^2 = \frac{1}{1 - \omega_L^2}, \quad X_\omega^2 U_{-\omega}^2 = \frac{c^2 \mathfrak{D}_L(\rho + \omega_L)}{1 + \omega_L} > Y_\xi^2 U_\xi^2.$$

Thus

$$(4.51) \quad q_{2L-1}(x_*, c)q_{2L-1}(x_*, -c) > 0.$$

At $x = 0$, the second-smallest singular value of A_{2L-1} is

$$s_{L,2}^{(0)} = \sqrt{1 + a^2 - 2a \cos \frac{\pi}{L}}.$$

Indeed, after separating odd and even coordinates, the two rectangular off-diagonal blocks have Gram matrices equal to the $(L-1)$ -dimensional tridiagonal Toeplitz matrix with diagonal $1 + a^2$ and off-diagonal a . Their singular values therefore have squares

$$1 + a^2 + 2a \cos \frac{k\pi}{L}, \quad 1 \leq k \leq L-1,$$

and each occurs twice, while the remaining singular value is zero. We claim

$$(4.52) \quad x_* + c < 2\sqrt{a} \sin \frac{\pi}{2L} \leq s_{L,2}^{(0)}.$$

Indeed, recall $\rho_{\text{ref}} = \cos \frac{\pi}{2L+1}$. Since $\rho > \rho_{\text{ref}}$ and U_{L-1} is strictly increasing above its largest zero,

$$u_- = U_{L-1}(\rho) > U_{L-1}(\rho_{\text{ref}}) =: u_{-, \text{ref}}, \quad u_{-, \text{ref}}^{-2} = 2(1 - \rho_{\text{ref}}).$$

The identities $c = a/u_-$ and $c^2 = 2a(\zeta - \rho)$ imply $\zeta - \rho < a(1 - \rho_{\text{ref}}) < 1 - \rho_{\text{ref}}$. Also

$$\frac{x_*^2}{2a} = \Delta_\omega \frac{\rho + \omega_L}{\zeta + \omega_L} < \Delta_\omega.$$

755 Put

$$756 \quad \theta_L = \frac{\pi}{4L}, \quad \Delta_L = \frac{\theta_L}{2L+1}.$$

757 Then $\Delta_\omega = 2 \sin^2 \theta_L$ and $1 - \rho_{\text{ref}} = 2 \sin^2(\theta_L - \Delta_L)$. Since all quantities are positive,
 758 the preceding quotient bound for x_*^2 gives

$$759 \quad x_* < 2\sqrt{a} \sin \theta_L.$$

760 Likewise,

$$761 \quad c^2 = 2a(\zeta - \rho) < 2a^2(1 - \rho_{\text{ref}}) = 4a^2 \sin^2(\theta_L - \Delta_L),$$

762 and hence

$$763 \quad c < 2a \sin(\theta_L - \Delta_L) < 2\sqrt{a} \sin(\theta_L - \Delta_L),$$

764 where the last inequality uses $0 < a < 1$, hence $a < \sqrt{a}$. It remains to prove

$$765 \quad (4.53) \quad \sin \theta_L + \sin(\theta_L - \Delta_L) < \sin(2\theta_L).$$

766 Now

$$767 \quad 2 \sin \theta_L - \sin(2\theta_L) = 2 \sin \theta_L (1 - \cos \theta_L) < \theta_L^3$$

768 and

$$769 \quad \sin \theta_L - \sin(\theta_L - \Delta_L) = \int_{\theta_L - \Delta_L}^{\theta_L} \cos u \, du > \Delta_L \cos \theta_L.$$

770 Thus (4.53) follows from $\Delta_L \cos \theta_L > \theta_L^3$, or $2 \cos \theta_L > \theta_L(\pi + 2\theta_L)$. For $0 < \theta_L \leq \pi/8$,
 771 using $\pi^2 < 10$,

$$772 \quad 2 \cos \theta_L > 2 - \theta_L^2 \geq \frac{59}{32} > \frac{25}{16} \geq \theta_L(\pi + 2\theta_L).$$

773 Consequently $x_* + c < 2\sqrt{a} \sin \frac{\pi}{2L}$. Finally,

$$774 \quad (s_{L,2}^{(0)})^2 - 4a \sin^2 \frac{\pi}{2L} = (1 - a)^2 \geq 0,$$

775 which proves (4.52).

776 Let $\mu_1, \dots, \mu_{2L-1}$ be the eigenvalues of the real symmetric matrix

$$777 \quad B_{2L-1}(x_*) = (x_* I_{2L-1} - A_{2L-1}) J_{2L-1}.$$

778 Right multiplication by J_{2L-1} preserves singular values, so the numbers $|\mu_i|$ are
 779 precisely the singular values of $x_* I_{2L-1} - A_{2L-1}$. Moreover,

$$780 \quad q_{2L-1}(x_*, c) q_{2L-1}(x_*, -c) = \prod_{i=1}^{2L-1} (\mu_i^2 - c^2).$$

781 The positive product in (4.51) is nonzero and therefore says that the number of indices
 782 with $|\mu_i| < c$ is even. On the other hand, Weyl's singular-value perturbation inequality
 783 and (4.52) give, for the second-smallest singular value s_{next} ,

$$784 \quad s_{\text{next}}(x_* I_{2L-1} - A_{2L-1}) \geq s_{L,2}^{(0)} - x_* > c,$$

785 so at most one index can satisfy $|\mu_i| < c$. The number is thus zero, and nonvanishing
 786 of the product excludes equality. Hence $s_{\min}(x_* I_{2L-1} - A_{2L-1}) > c_L$. Since $0 <$

787 $x_* < 2r \sin \frac{\pi}{2L}$, the point lies strictly in the positive central gap, and $d_{L-1} > c_L$. This
 788 completes the proof of (4.36).

789 *Completion of the comparison.* We finish the index comparison. Put $M = N + 1$,
 790 so the larger and smaller matrices have orders $M - 1$ and $N - 1$, respectively. If
 791 $\Sigma = \text{diag}(1, -1, 1, -1, \dots)$, then $\Sigma A_n \Sigma = -A_n$, so reflection reduces every noncentral
 792 comparison to a positive gap. By the interiority observation following the definition of
 793 the gap heights, choose $x_j \in G_{M-1,j}^\circ$ at which its height is attained. The hyperbolic or
 794 elliptic construction above produces a point $\hat{x}_j \in G_{N-1,j}$, and chord exhaustion gives

$$795 \quad \gamma_{N-1,j} \geq g_{N-1}(\hat{x}_j) > g_{M-1}(x_j) = \gamma_{M-1,j}.$$

796 The reflected inequality handles the corresponding negative gap.

797 If $M = 2\ell_* + 1$, the remaining single central height of the larger, even-order matrix
 798 is c_{ℓ_*} , whereas each central height of the smaller odd-order matrix is $d_{\ell_*-1} > c_{\ell_*}$. If
 799 $M = 2\ell_* + 2$, the two central heights of the larger odd-order matrix equal $d_{\ell_*} < c_{\ell_*}$,
 800 where c_{ℓ_*} is the single central height of the smaller even-order matrix. Thus every
 801 height of the larger problem is strictly smaller than at least one height of the smaller
 802 problem. Choose j_* with $\gamma_{M-1,j_*} = \max_j \gamma_{M-1,j}$. Then

$$803 \quad \gamma_{M-1} = \gamma_{M-1,j_*} < \max_k \gamma_{N-1,k} = \gamma_{N-1}.$$

804 Since $M - 1 = n + 1$ and $N - 1 = n$, this proves $\gamma_{n+1} < \gamma_n$. $\square$

805 **5. Topology, quantitative bounds, and the transition scale.** We begin
 806 with the standard component property that reduces the planar topology to the real
 807 spectral interval.

808 LEMMA 5.1. *Every connected component of $\sigma_\varepsilon(A_n)$ contains an eigenvalue of A_n ;*
 809 *this standard component property holds for arbitrary matrices [9].*

810 *Proof.* The estimate

$$811 \quad s_{\min}(zI_n - A_n) \geq |z| - \|A_n\|_2$$

812 shows that the pseudospectrum is bounded. Let $\mathcal{C}$ be one of its components. If
 813 an eigenvalue λ belonged to $\overline{\mathcal{C}}$, continuity of the least singular value would give a
 814 ball about λ contained in the pseudospectrum. That ball meets $\mathcal{C}$, so maximality of
 815 connected components places λ in $\mathcal{C}$.

816 Suppose that $\mathcal{C}$ contains no eigenvalue. Its compact closure is then disjoint from
 817 the spectrum. On a neighbourhood of $\overline{\mathcal{C}}$, the function $u(z) = \|(zI_n - A_n)^{-1}\|_2$ is
 818 continuous and subharmonic; see [26]. Moreover, $u \leq \varepsilon^{-1}$ on $\partial\mathcal{C}$, since a strict
 819 inequality would put a neighbourhood of that boundary point in the same component.
 820 The maximum principle gives $u \leq \varepsilon^{-1}$ throughout $\mathcal{C}$, contrary to the definition of the
 821 pseudospectrum. $\square$

822 The diagonal gauge introduced in (1.4) gives

$$823 \quad A_n = \mathcal{G}_n(r) [r(V_n + V_n^T)] \mathcal{G}_n(r)^{-1},$$

824 and hence

$$825 \quad (5.1) \quad \sigma(A_n) = \left\{ 2r \cos \frac{k\pi}{n+1} : 1 \leq k \leq n \right\}.$$

826 Put

$$827 \quad \Omega = \sigma_\varepsilon(A_n), \quad \Omega_{\mathbb{R}} = \Omega \cap \mathbb{R} = \{x \in \mathbb{R} : g_n(x) < \varepsilon\}.$$

828 Because A_n is real, the least singular value is invariant under complex conjugation.
829 Lemma 3.1 therefore implies that

$$830 \quad \mathcal{H}_t(x + iy) = x + i(1 - t)y, \quad 0 \leq t \leq 1,$$

831 maps Ω into itself. If $\mathcal{C}$ is a component of Ω , the path $t \mapsto \mathcal{H}_t(z)$ starts in $\mathcal{C}$ and hence
832 remains there. Thus $\mathcal{H}_t$ retracts $\mathcal{C}$ onto $\mathcal{C} \cap \mathbb{R}$.

833 By Lemma 5.1, this real intersection is nonempty. It is connected, being the
834 continuous image $\mathcal{H}_1(\mathcal{C})$, and hence lies in one interval component $\mathcal{K}$ of $\Omega_{\mathbb{R}}$. Conversely,
835 $\mathcal{K}$ is a connected subset of Ω meeting $\mathcal{C}$, so $\mathcal{K} \subset \mathcal{C}$. Therefore $\mathcal{C} \cap \mathbb{R} = \mathcal{K}$, and the
836 components of Ω correspond bijectively to the interval components of $\Omega_{\mathbb{R}}$. In particular,
837 each $\mathcal{C}$ retracts onto an interval and is contractible.

838 If Ω is connected, its real intersection is an interval containing every eigenvalue,
839 and therefore contains their convex hull I_n^{sp} . Conversely, if $I_n^{\text{sp}} \subset \Omega$, that interval
840 belongs to one component and contains every eigenvalue; Lemma 5.1 rules out any
841 other component. Compactness of I_n^{sp} now gives

$$842 \quad (5.2) \quad \Omega \text{ is connected} \iff I_n^{\text{sp}} \subset \Omega \iff \max_{x \in I_n^{\text{sp}}} g_n(x) < \varepsilon \iff \gamma_n < \varepsilon.$$

843 The strict inequality reflects the open definition of the pseudospectrum; equality leaves
844 at least one real bridge point outside.

845 We now give quantitative estimates. Put $N = n + 1$ and recall $\mathcal{L}_n(a)$ and $\mathcal{U}_n(a)$
846 from (2.3)–(2.2).

847 PROPOSITION 5.2 (Sharp-order gap bounds). *For every $n \geq 2$,*

$$848 \quad (5.3) \quad \mathcal{L}_n(a) \leq \gamma_n \leq \mathcal{U}_n(a).$$

849 *In particular, both bounds have order $(n + 1)a^{n/2}$, with constants depending only on a .*

850 *Proof.* For $x = 2r \cos \theta \in I_n^{\text{sp}}$, set

$$851 \quad \mathbf{v} = (v_j)_{j=1}^n, \quad v_j = r^{j-1} \sin(j\theta), \quad 1 \leq j \leq n.$$

852 All entries of $(xI_n - A_n)\mathbf{v}$, except the last, vanish, and the last one is $r^n \sin(N\theta)$. Thus

$$853 \quad g_n(2r \cos \theta) \leq \frac{r^n |\sin(N\theta)|}{\left(\sum_{j=1}^n r^{2j-2} \sin^2(j\theta)\right)^{1/2}} \leq r^n \frac{|\sin(N\theta)|}{\sin \theta}.$$

854 Since $\pi/N \leq \theta \leq (N - 1)\pi/N$, this proves the upper bound.

855 For the lower bound, put $p_\theta = re^{i\theta}$, $q_\theta = re^{-i\theta}$, so that $p_\theta + q_\theta = x$ and $p_\theta q_\theta = a$.
856 For $\mathbf{u} = (u_j)_{j=1}^n \in \mathbb{C}^n$, put $u_0 = u_{n+1} = 0$, and define maps from $\mathbb{C}^n$ to $\mathbb{C}^{n+1}$ by

$$857 \quad (\mathbf{C}_\theta \mathbf{u})_j = p_\theta u_j - u_{j+1}, \quad (\mathbf{D}_\theta \mathbf{u})_j = u_j - \overline{q_\theta} u_{j+1}, \quad 0 \leq j \leq n.$$

858 For $1 \leq k \leq n$, direct multiplication gives

$$859 \quad (\mathbf{D}_\theta^* \mathbf{C}_\theta \mathbf{u})_k = (\mathbf{C}_\theta \mathbf{u})_k - q_\theta (\mathbf{C}_\theta \mathbf{u})_{k-1} = (p_\theta + q_\theta)u_k - u_{k+1} - p_\theta q_\theta u_{k-1}.$$

860 Hence

$$861 \quad (5.4) \quad xI_n - A_n = \mathbf{D}_\theta^* \mathbf{C}_\theta.$$

The two Gram matrices are diagonally unitarily similar to the tridiagonal Toeplitz matrix with diagonal $1 + r^2$ and off-diagonal $-r$. Its eigenvalues are $1 + r^2 - 2r \cos \frac{k\pi}{N}$, $1 \leq k \leq n$, so

$$(5.5) \quad \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right)^{1/2}.$$

Put

$$\mathcal{S}_N = \sum_{j=0}^n r^{2j} = \frac{1 - r^{2N}}{1 - r^2}.$$

Unit normal vectors to the two hyperplane ranges are

$$\mathbf{u}_C = \mathcal{S}_N^{-1/2}(\overline{p_\theta}^n, \overline{p_\theta}^{n-1}, \dots, 1)^T, \quad \mathbf{u}_D = \mathcal{S}_N^{-1/2}(1, q_\theta, \dots, q_\theta^n)^T.$$

In the polar decompositions

$$\mathbf{C}_\theta = Q_C(\mathbf{C}_\theta^* \mathbf{C}_\theta)^{1/2}, \quad \mathbf{D}_\theta = Q_D(\mathbf{D}_\theta^* \mathbf{D}_\theta)^{1/2},$$

the matrices Q_C, Q_D are isometries onto these ranges. Two hyperplanes in $\mathbb{C}^{n+1}$ intersect in dimension $n - 1$, so the cross-Gram matrix $Q_D^* Q_C$ has $n - 1$ singular values equal to 1; the remaining one is the cosine of the remaining principal angle, $|\langle \mathbf{u}_D, \mathbf{u}_C \rangle|$. The geometric sum gives

$$(5.6) \quad \begin{aligned} |\langle \mathbf{u}_D, \mathbf{u}_C \rangle| &= \frac{r^n}{\mathcal{S}_N} \left| \sum_{j=0}^{N-1} e^{i(2j-(N-1))\theta} \right| \\ &= \frac{r^n(1 - r^2)}{1 - r^{2N}} \left| \frac{\sin(N\theta)}{\sin \theta} \right| =: \chi_N(\theta). \end{aligned}$$

Finally,

$$xI_n - A_n = (\mathbf{D}_\theta^* \mathbf{D}_\theta)^{1/2} Q_D^* Q_C (\mathbf{C}_\theta^* \mathbf{C}_\theta)^{1/2}.$$

Using $s_{\min}(XYZ) \geq s_{\min}(X)s_{\min}(Y)s_{\min}(Z)$ and (5.5) gives

$$g_n(2r \cos \theta) \geq \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right) \chi_N(\theta).$$

Putting $\theta = 3\pi/(2N)$, which lies in the first spectral gap, proves the lower bound.

Finally,

$$\mathcal{U}_n(a) \leq \frac{1}{2}(n+1)r^n,$$

while

$$(5.7) \quad \mathcal{L}_n(a) \geq C_{\text{lb}}(a)(n+1)r^n, \quad C_{\text{lb}}(a) = \frac{2(1-r)^2(1-r^2)}{3\pi}.$$

These follow from $\sin \frac{\pi}{N} \geq 2/N$, $1 + r^2 - 2r \cos \frac{\pi}{N} \geq (1-r)^2$, and $\sin \frac{3\pi}{2N} \leq 3\pi/(2N)$. This proves the asserted sharp order. $\square$

Proof of Theorem 2.1. By Proposition 5.2, $\gamma_n \leq \mathcal{U}_n(a) \rightarrow 0$. Lemma 4.1 says that γ_n is strictly decreasing. Equation (5.2) therefore shows that, once the pseudospectrum is connected, it stays connected in every larger dimension. This proves $N_e = N_f$. Contractibility of every component was proved immediately before (5.2).

893 If $\mathcal{L}_n(a) \geq \varepsilon$, then $\gamma_n \geq \varepsilon$, so the pseudospectrum in dimension n is disconnected.
 894 If $\mathcal{U}_n(a) < \varepsilon$, then $\gamma_n < \varepsilon$, so it is connected. Strict decrease of γ_n now gives (2.4).
 895 The scale in (2.4) can be displayed in closed form. Put

$$896 \quad \Theta_n = (n+1)r^n, \quad \mathcal{T}_r(t) = \min\{N \geq 2 : \Theta_m < t \text{ for every } m \geq N\}.$$

897 Equations (5.7) and $\mathcal{U}_n(a) \leq \Theta_n/2$ give

$$898 \quad (5.8) \quad \boxed{\mathcal{T}_r(\varepsilon/C_{\text{lb}}(a)) \leq N_c(a, \varepsilon) \leq \mathcal{T}_r(2\varepsilon).}$$

899 Indeed, (5.2) gives

$$900 \quad N_c(a, \varepsilon) = \min\{N \geq 2 : \gamma_m < \varepsilon \text{ for every } m \geq N\}.$$

901 The lower estimate $C_{\text{lb}}(a)\Theta_m \leq \gamma_m$ forces $\Theta_m < \varepsilon/C_{\text{lb}}(a)$ throughout the tail begin-
 902 ning at N_c . Conversely, $\Theta_m < 2\varepsilon$ for every $m \geq \mathcal{T}_r(2\varepsilon)$, together with $\gamma_m \leq \Theta_m/2$,
 903 forces connectedness throughout that tail. Thus the polynomial factor $n+1$, as well
 904 as the exponential factor $a^{n/2}$, is sharp on both sides.

905 For completeness, let

$$906 \quad \beta = -\log r = \frac{1}{2} \log(1/a).$$

907 For $0 < t < 3r^2 = \Theta_2$, let $x_t > 2$ be the solution on the decreasing branch of
 908 $(x+1)e^{-\beta x} = t$. This branch exists because

$$909 \quad 0 < \beta r t < 3\beta r^3 \leq e^{-1};$$

910 the last inequality is $ue^{-u} \leq e^{-1}$ with $u = 3\beta$. Moreover, $t < \Theta_2$ places the
 911 descending root strictly beyond $x = 2$. Putting $y_t = x_t + 1$, the equation becomes
 912 $(-\beta y_t)e^{-\beta y_t} = -\beta r t$, whence $y_t = -\beta^{-1}W_{-1}(-\beta r t)$. Therefore

$$913 \quad (5.9) \quad \mathcal{T}_r(t) = \left\lfloor -\frac{1}{\beta} W_{-1}(-\beta r t) \right\rfloor,$$

914 where W_{-1} is the lower real branch of the Lambert W -function. Indeed, the continuous
 915 function $(x+1)r^x$ is unimodal and, after its descending root, is strictly below t . Hence
 916 the integer tail condition is $N > x_t$; its least solution is $\lfloor x_t \rfloor + 1 = \lfloor x_t + 1 \rfloor$, including
 917 when x_t is an integer.

918 The standard lower-branch expansion [11]

$$919 \quad -W_{-1}(-u) = \log(1/u) + \log \log(1/u) + O(1), \quad u \downarrow 0,$$

920 combined with (5.8) proves, for each fixed $a \in (0, 1)$, as $\varepsilon \downarrow 0$,

$$921 \quad (5.10) \quad \boxed{N_c(a, \varepsilon) = \frac{2}{\log(1/a)} (\log(1/\varepsilon) + \log \log(1/\varepsilon)) + O_a(1).}$$

922 The remainder is not asserted to be uniform as $a \uparrow 1$. Finally, for $|x| \leq r$,

$$923 \quad \det((xI_2 - A_2)^*(xI_2 - A_2) - a^2 I_2) = x^2(x^2 - 2a - 2a^2) \leq 0.$$

924 The matrix inside the determinant is Hermitian, and the nonpositive product of its
 925 two eigenvalues forces its least eigenvalue to be nonpositive. Thus $s_{\min}(xI_2 - A_2) \leq a$.
 926 At $x = 0$, the singular values of $-A_2$ are 1 and a , so equality is attained and $\gamma_2 = a$.
 927 Hence, by (5.2),

$$928 \quad (5.11) \quad N_c(a, \varepsilon) = 2 \iff \varepsilon > a.$$

929 This proves all assertions. □

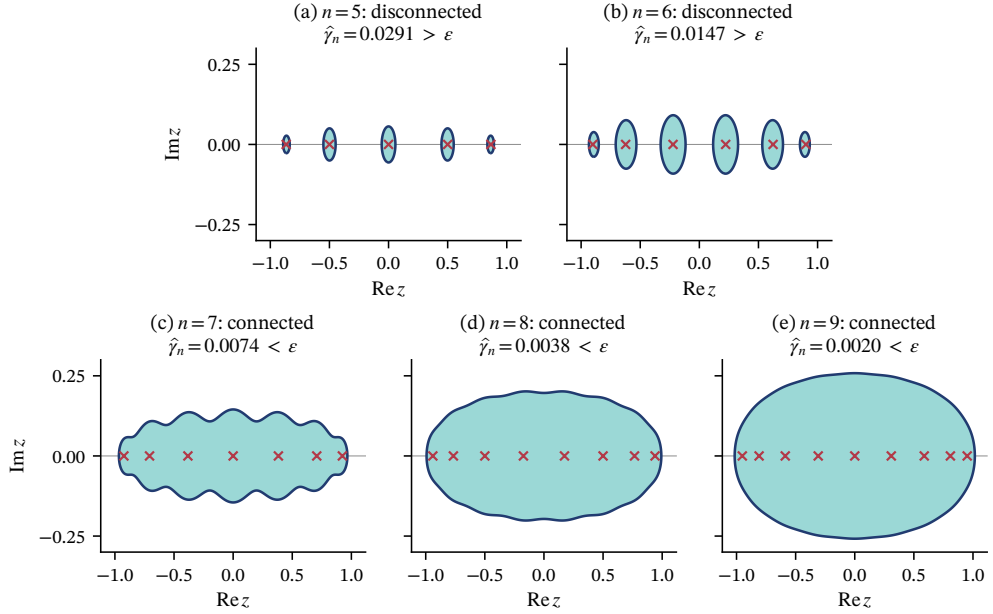


FIG. 1. Topology change indicated by the float64 computation at $a = 1/4$ and $\varepsilon = 10^{-2}$. Shading depicts $\{z : s_{\min}(zI_n - A_n) < \varepsilon\}$, and crosses are open-boundary eigenvalues. Gapwise SVD estimates give $(\hat{\gamma}_5, \dots, \hat{\gamma}_9) = (0.02909836, 0.01465829, 0.007364219, 0.003820033, 0.001992747)$. Applying the grid/Lipschitz check to the float64 samples places the computed margins for $n = 5, 6$ above ε and those for $n = 7, 8, 9$ below it, yielding the reported numerical threshold $N_c = 7$. As explained in the text, these checks are not interval certificates. The merger is a perturbation-level-dependent pseudospectral event, not the onset of skin localization.

6. Numerical illustrations and reproducibility. The computations in this section illustrate the theorem but are not used in its proof. The complete script `make_hatano_nelson_figures.py`, supplied with the manuscript, uses NumPy, SciPy, and Matplotlib. It constructs $A_n(a)$ directly, computes Cartesian pseudospectral contours by singular value decomposition, and obtains gapwise estimates of γ_n by bounded scalar optimization. As a separate check, let M_h be the largest exact least singular value sampled on uniform real-gap grids of maximum spacing h . The 1-Lipschitz property of $x \mapsto s_{\min}(xI_n - A_n)$ gives the rigorous implication

$$M_h \leq \gamma_n \leq M_h + \frac{h}{2}$$

for exact samples. The supplied implementation applies this test to NumPy/SciPy float64 SVD values, but it does not enclose SVD or eigenvalue roundoff. Its topology labels are therefore reproducible floating-point checks, conditional on the computed samples, rather than validated interval certificates. They avoid reliance on pixel connectivity or the scalar optimizer and are unused in every proof.

The five panels of Figure 1 illustrate the computed classification: two disconnected cases, the first connected case, and two subsequent connected cases. As proved above, whenever the true pseudospectrum is connected it has no hole: every vertical section is an interval, and the real trace contains the full open-boundary spectral segment.

Figure 2 separates the two roles of the analysis. The strict comparison theorem

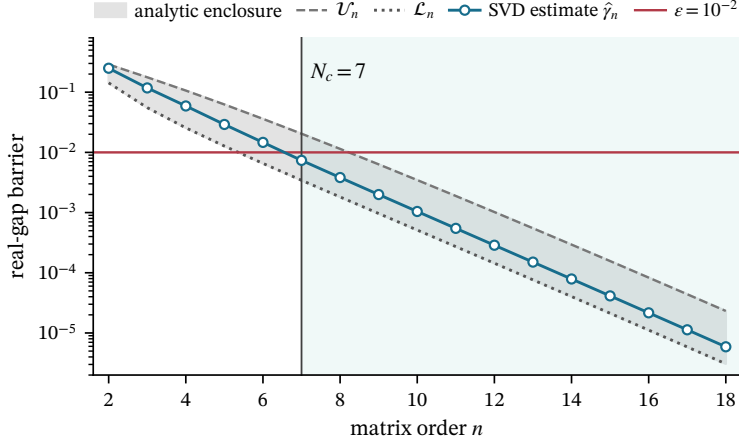


FIG. 2. Gapwise float64 SVD estimates $\hat{\gamma}_n$ of the strictly decreasing barriers and the exact analytic enclosure $\mathcal{L}_n(a) \leq \gamma_n \leq \mathcal{U}_n(a)$ for $a = 1/4$. The plotted estimates first cross below $\varepsilon = 10^{-2}$ at $n = 7$, again indicating the reported numerical threshold. Independently, the exact analytic bounds in (2.4) give $6 \leq N_c \leq 9$. The nearly parallel log-scale curves display the common $(n+1)a^{n/2}$ scale.

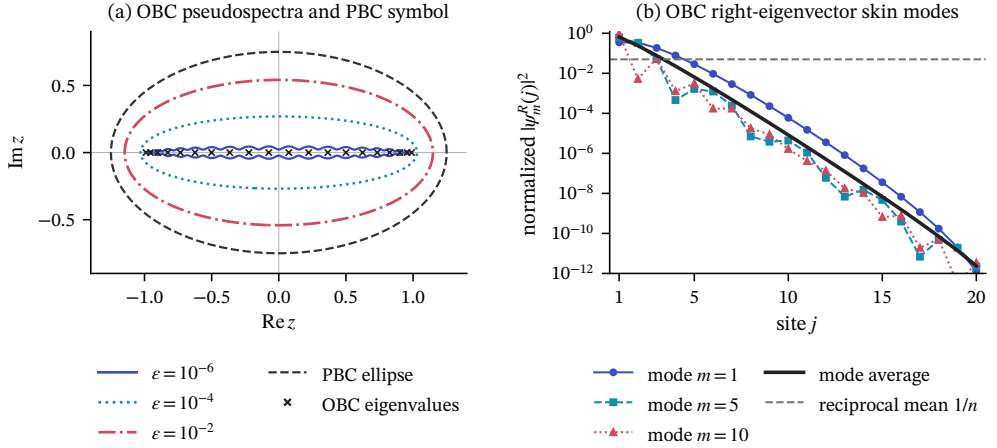


FIG. 3. Hatano–Nelson interpretation for $a = 1/4$ and $n = 20$. (a) The open-boundary eigenvalues lie in the real segment $[-2\sqrt{a}, 2\sqrt{a}]$, whereas the periodic symbol $e^{ik} + ae^{-ik}$ traces an ellipse. The nested contours are boundaries of strict pseudospectra. (b) Right eigenvectors $\psi_m^R(j) \propto a^{(j-1)/2} \sin \frac{m\pi j}{n+1}$ accumulate at the left edge. Reversing the site convention reverses the edge but not the pseudospectral conclusions.

identifies a unique crossing of a fixed ε , while the elementary upper and lower envelopes prove its sharp exponential-polynomial scale without numerically resolving the maximizing gap.

Figure 3 displays the two complementary manifestations of the imaginary gauge: the periodic/open spectral contrast and the open-boundary skin profile. Although all open-boundary eigenvalues are real, the displayed pseudospectral contours elongate rapidly toward the periodic ellipse. The ellipse is a reference curve, not a large-level limit: for sufficiently large perturbation levels the pseudospectra extend beyond it.

7. Hatano–Nelson interpretation and conclusions. To connect the dimensionless matrix to the usual second-quantized notation, consider

$$\hat{H}_{\text{OBC}} = - \sum_{j=1}^{n-1} \left(t_L \hat{c}_j^\dagger \hat{c}_{j+1} + t_R \hat{c}_{j+1}^\dagger \hat{c}_j \right), \quad t_L > t_R > 0.$$

Its one-particle matrix is $-t_L A_n(a)$, $a = t_R/t_L$. The conventional minus sign is immaterial here because $\Sigma A_n(a) \Sigma = -A_n(a)$ for $\Sigma = \text{diag}(1, -1, \dots, (-1)^{n-1})$. Put

$$t_0 = \sqrt{t_L t_R}, \quad h = \frac{1}{2} \log(t_L/t_R) = \frac{1}{2} \log(1/a).$$

For the unitarily equivalent positive-sign representative $t_L A_n(a)$, multiplying the dimensionless relation (1.4) by t_L gives the eigenvalues and unnormalized right and left modes

$$E_m = 2t_0 \cos \frac{m\pi}{n+1},$$

$$\psi_m^R(j) = e^{-h(j-1)} \sin \frac{j m \pi}{n+1},$$

$$\psi_m^L(j) = e^{h(j-1)} \sin \frac{j m \pi}{n+1}, \quad 1 \leq j, m \leq n.$$

These mode profiles are closely related to the original Hatano–Nelson eigenfunction analysis [14]. Thus the right-mode amplitude skin depth is

$$\ell_{\text{skin}} = h^{-1} = \frac{2}{\log(1/a)},$$

whereas the gauge condition number is $\kappa_2(\mathcal{G}_n(e^{-h})) = e^{h(n-1)} = a^{-(n-1)/2}$. The quantitative bounds in Proposition 5.2 give, more precisely,

$$\gamma_n^{-1} \asymp_a \frac{e^{hn}}{n+1}.$$

Thus gauge conditioning and worst-case spectral sensitivity have the same exponential rate h , up to fixed- a constants and the displayed polynomial factor.

The perturbational characterization following (1.1) clarifies the meaning of connectedness. It says that the *union* of eigenvalue locations attainable under all unstructured perturbations of norm below ε has merged into one uncertainty region; it does not identify a single perturbation whose eigenvalues populate all of the connecting bridges. Equivalently, for $A \in \mathbb{C}^{n \times n}$ and $z \notin \sigma(A)$,

$$\|(zI_n - A)^{-1}\|_2 = \max_{\mathbf{f} \neq 0} \frac{\|\mathbf{u}\|_2}{\|\mathbf{f}\|_2}, \quad (zI_n - A)\mathbf{u} = \mathbf{f},$$

so the pseudospectrum is the region of worst-case resolvent gain larger than ε^{-1} ; at a spectral point the gain is interpreted as infinite. The quantity γ_n is the hardest real-axis bridge between adjacent open-boundary spectral islands.

For the dimensional Hamiltonian $t_L A_n(a)$, the merger level is $t_L \gamma_n$. If $\varepsilon_{\text{phys}}$ is a perturbation level with energy units and $\delta = \varepsilon_{\text{phys}}/t_L$ is its dimensionless strength, then (2.5) becomes

$$N_c(a, \delta) = \ell_{\text{skin}} (\log(1/\delta) + \log \log(1/\delta)) + O_a(1).$$

The leading coefficient is exactly the amplitude skin depth. At fixed dimensionless δ , its leading small- δ value decreases as a decreases; this is an asymptotic comparison, not a claim that N_c is monotone in a at every finite perturbation level. The limit $a \uparrow 1$ is not uniform: at $a = 1$ the matrix is normal and the connectedness scale is governed by ordinary Hermitian eigenvalue spacing. Likewise, restricting perturbations to onsite, hopping, boundary, or Toeplitz structure can produce a smaller attainable set and requires a structured-pseudospectral analysis. Toeplitz-structured pseudospectra are treated in [7] and [10]. Robustness under random imperfections is studied in [3]; pseudospectral boundary sensitivity in correlated or disordered systems in [20]; and complex and structured-pseudospectral scaling in asymmetric lattices in [15].

In summary, the planar topology is controlled by a one-dimensional barrier sequence: vertical monotonicity makes every component contractible, persymmetry and folded Chebyshev coordinates make that sequence strictly decreasing with size, and rectangular factorizations identify its sharp scale. For the clean open Hatano–Nelson chain this provides an exact finite-size merger criterion, while keeping distinct the skin effect itself, worst-case spectral uncertainty, and perturbation-structured physical response.

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